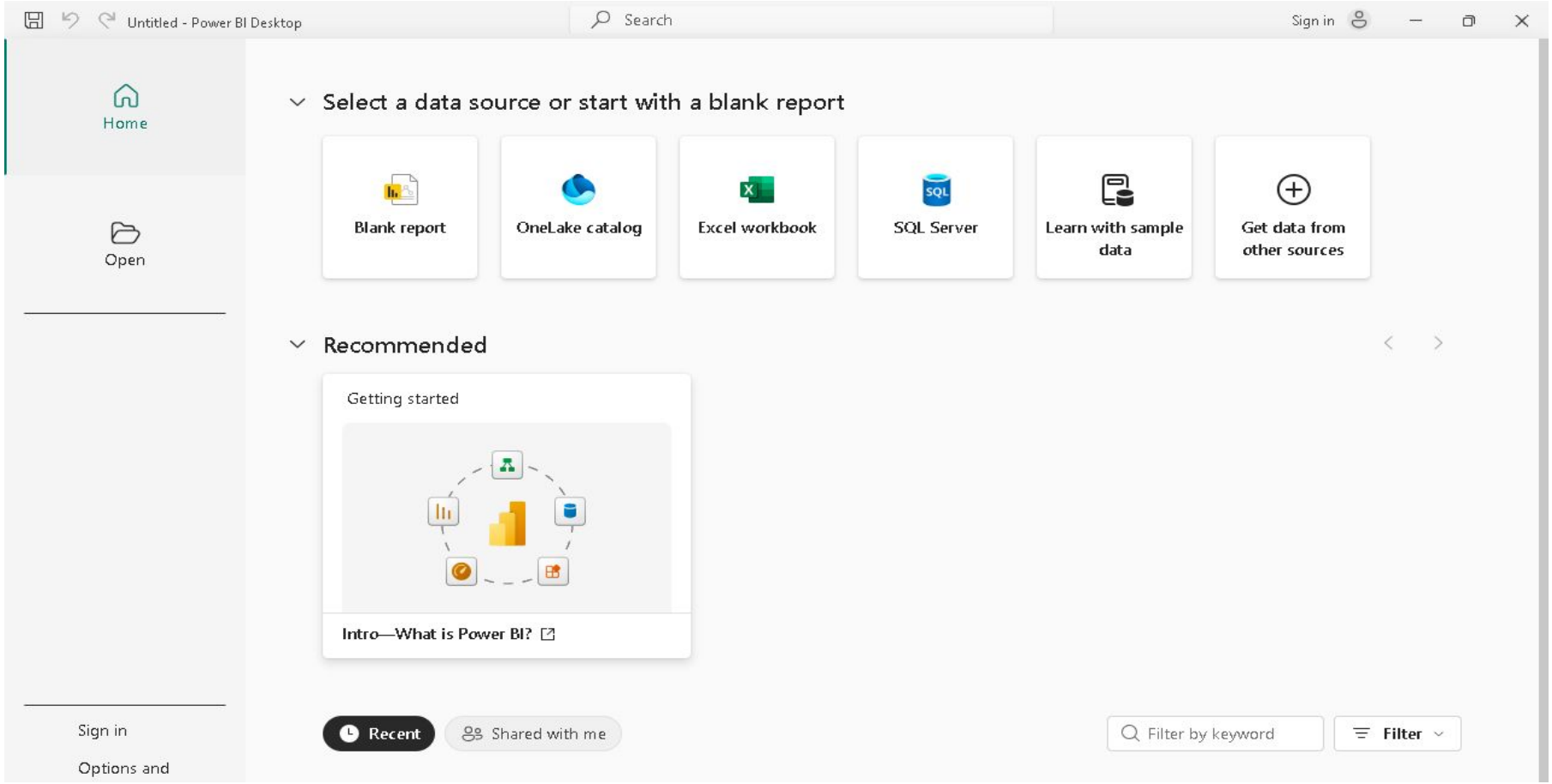
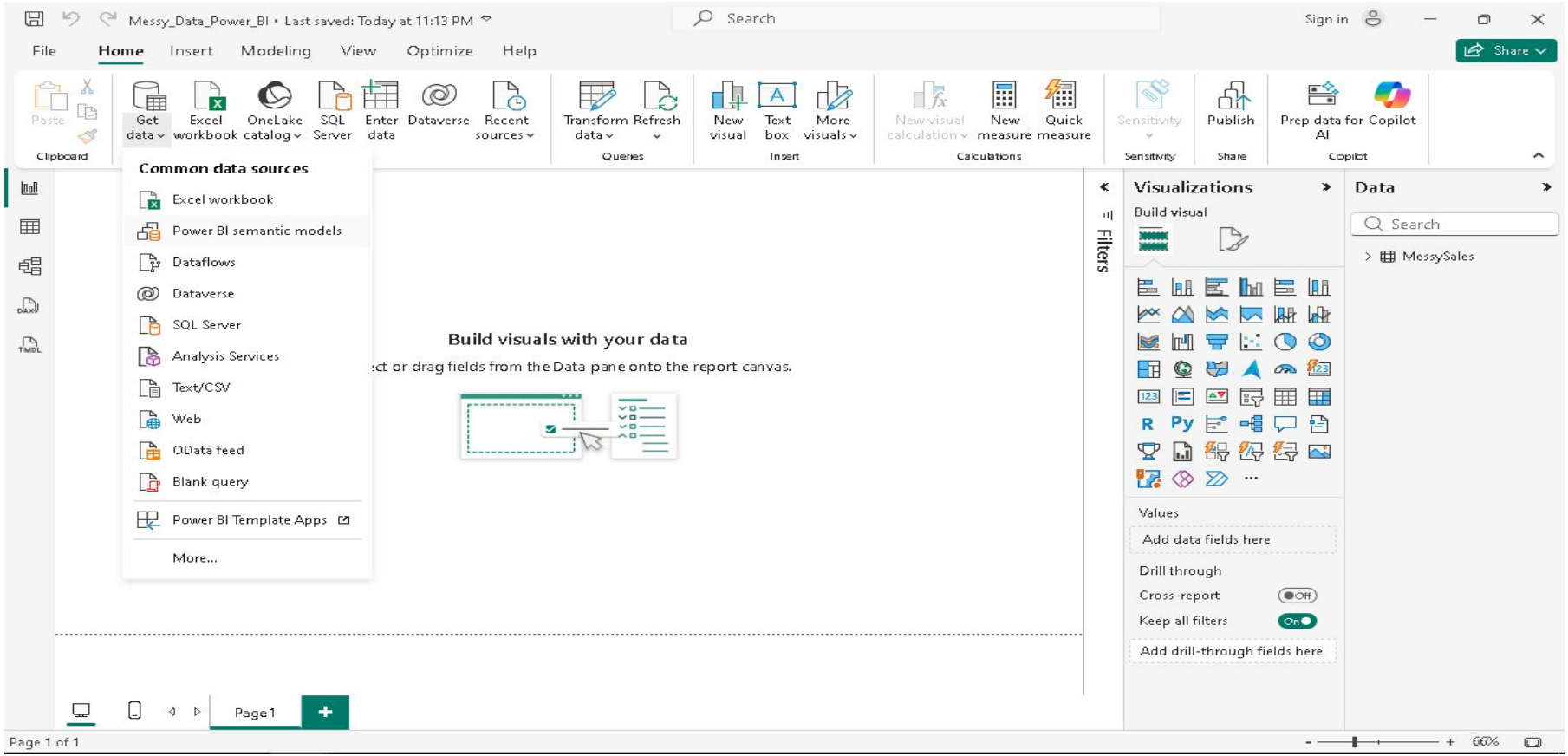
End-to-End Data Analytics in Power BI: From Data Cleaning to DAX Insights

Kenneth A. Camasura
Instructor: Joseph A. Vistal
IS 107 - Business Intelligence

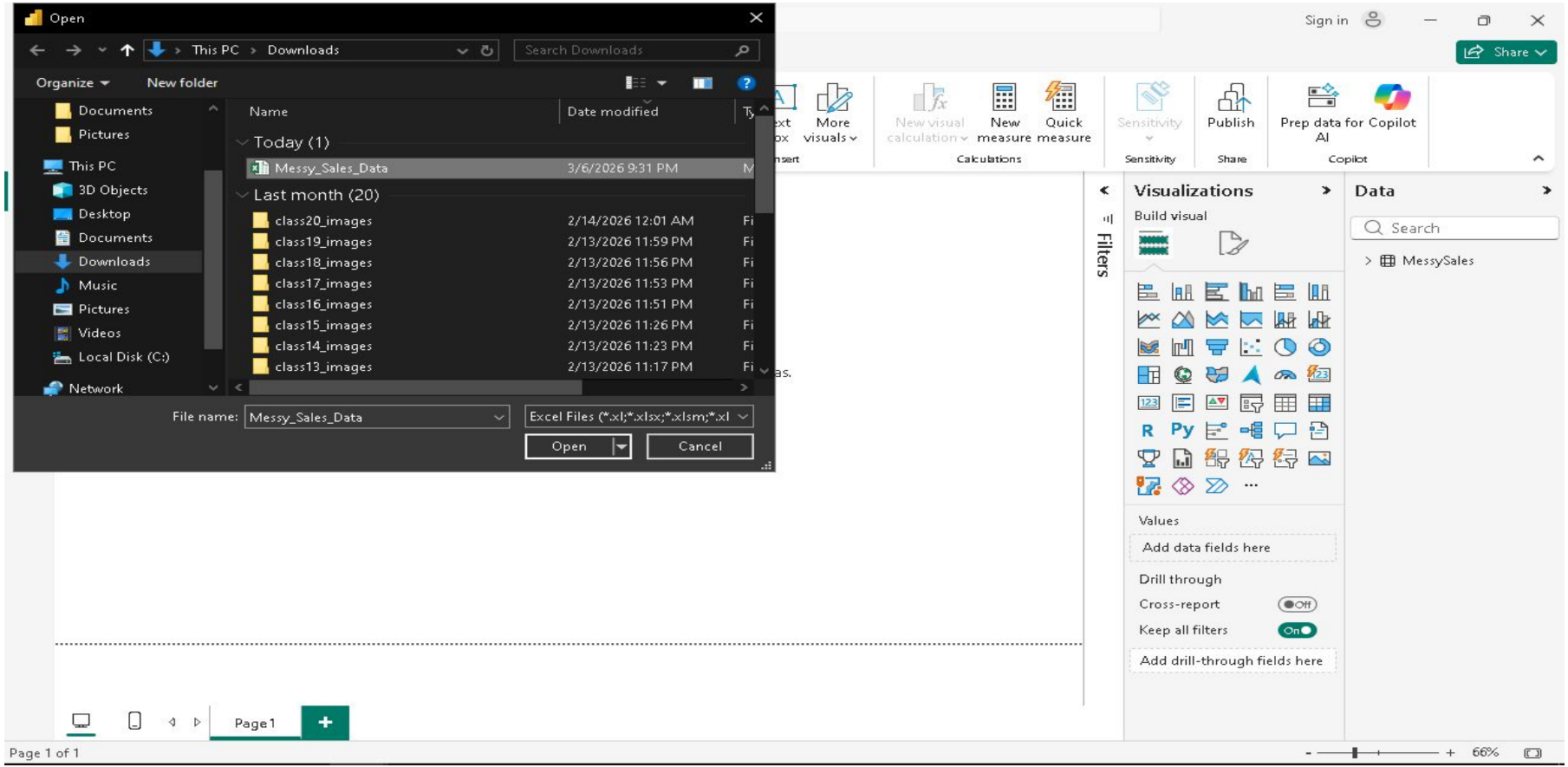
Power Query – Data Cleaning Part 1 | Step 1. Open Power BI Desktop.



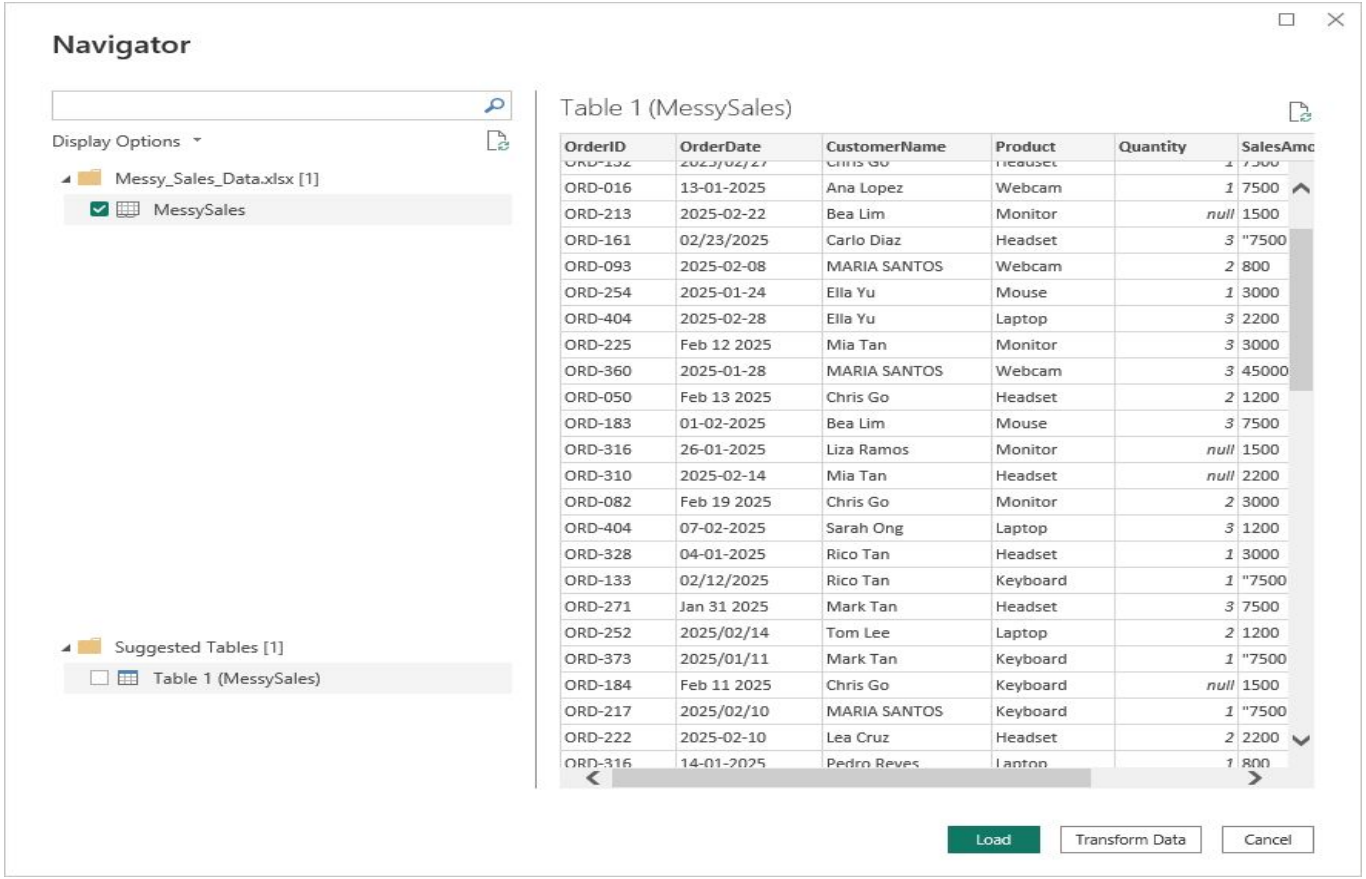
Power Query – Data Cleaning Part 1 | Step 2. Click Home → Get Data → Excel Workbook.



Power Query – Data Cleaning Part 1 | Step 3. Select Messy_Sales_Data.xlsx.

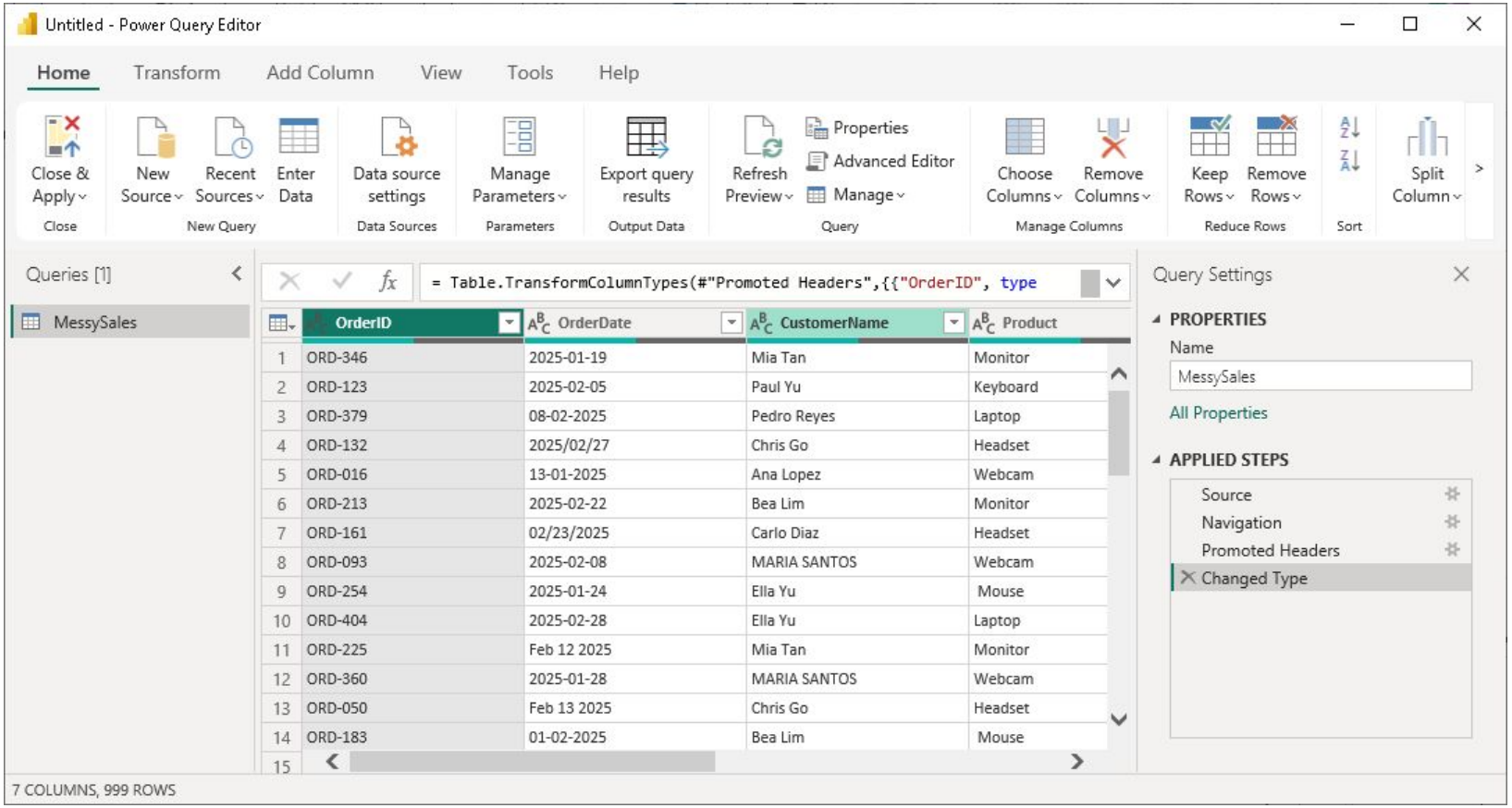


Power Query – Data Cleaning Part 1 | Step 4. Click Transform Data to open Power Query Editor.

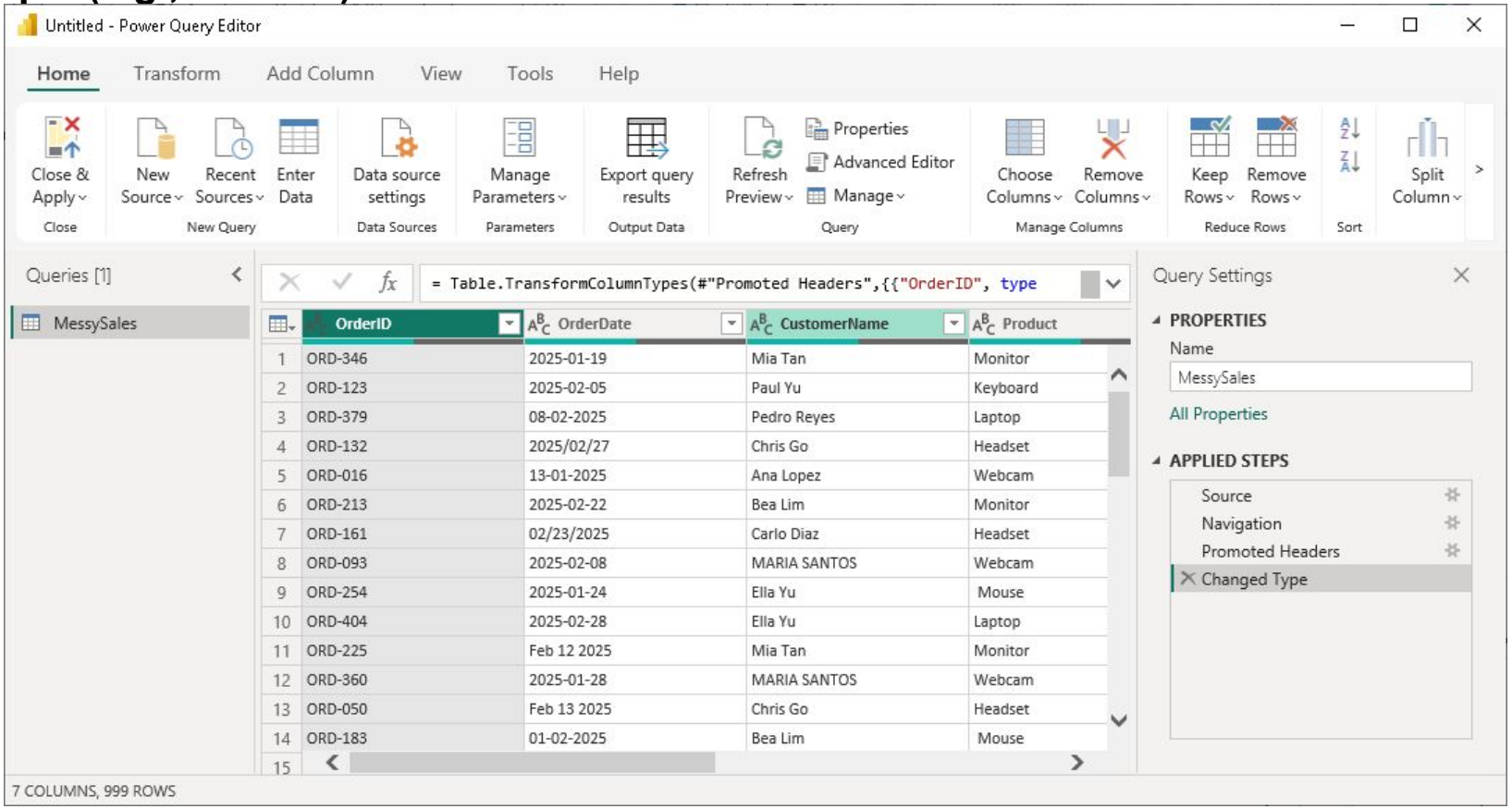


OrderID	OrderDate	CustomerName	Product	Quantity	SalesAmo
ORD-016	13-01-2025	Ana Lopez	Webcam	1	7500
ORD-213	2025-02-22	Bea Lim	Monitor	null	1500
ORD-161	02/23/2025	Carlo Diaz	Headset	3	"7500
ORD-093	2025-02-08	MARIA SANTOS	Webcam	2	800
ORD-254	2025-01-24	Ella Yu	Mouse	1	3000
ORD-404	2025-02-28	Ella Yu	Laptop	3	2200
ORD-225	Feb 12 2025	Mia Tan	Monitor	3	3000
ORD-360	2025-01-28	MARIA SANTOS	Webcam	3	45000
ORD-050	Feb 13 2025	Chris Go	Headset	2	1200
ORD-183	01-02-2025	Bea Lim	Mouse	3	7500
ORD-316	26-01-2025	Liza Ramos	Monitor	null	1500
ORD-310	2025-02-14	Mia Tan	Headset	null	2200
ORD-082	Feb 19 2025	Chris Go	Monitor	2	3000
ORD-404	07-02-2025	Sarah Ong	Laptop	3	1200
ORD-328	04-01-2025	Rico Tan	Headset	1	3000
ORD-133	02/12/2025	Rico Tan	Keyboard	1	"7500
ORD-271	Jan 31 2025	Mark Tan	Headset	3	7500
ORD-252	2025/02/14	Tom Lee	Laptop	2	1200
ORD-373	2025/01/11	Mark Tan	Keyboard	1	"7500
ORD-184	Feb 11 2025	Chris Go	Keyboard	null	1500
ORD-217	2025/02/10	MARIA SANTOS	Keyboard	1	"7500
ORD-222	2025-02-10	Lea Cruz	Headset	2	2200
ORD-316	14-01-2025	Pedro Reyes	Laptop	1	800

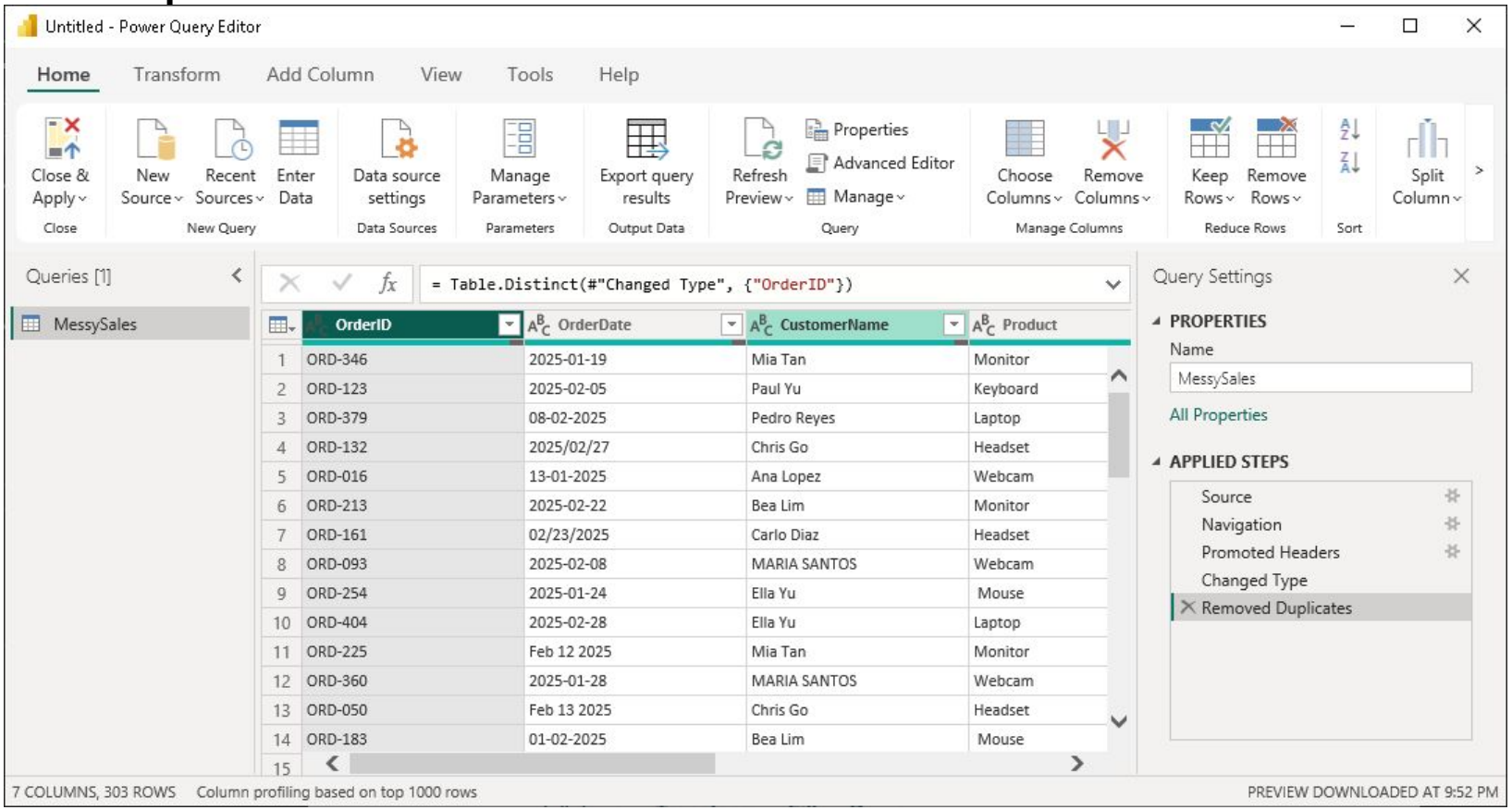
Power Query – Data Cleaning Part 2 | Step 1.Select the table.



Power Query – Data Cleaning Part 2 | Step 2. Highlight the column(s) that should be unique (e.g., OrderID).



Power Query – Data Cleaning Part 2 | Step 3. Click Home → Remove Rows → Remove Duplicates.



Power Query – Data Cleaning Part 3 | Step 1. Identify columns with null values.

Messy_Data_Power_BI

Home | Transform | Add Column | View | Tools | Help

Close & Apply | New Source | Recent Sources | Enter Data | Data source settings | Manage Parameters | Export query results | Refresh Preview | Properties | Advanced Editor | Manage | Choose Columns | Remove Columns | Keep Rows | Remove Rows | Split Column

Close | New Query | Data Sources | Parameters | Output Data | Query | Manage Columns | Reduce Rows | Sort

Queries [2]

- MessySales
- MessySales (2)

```
= Table.TransformColumnTypes(#"Promoted Headers",{{"OrderID", type
```

		Quantity	SalesAmount	Region
125		2	1500	Luzon
126		null	7500	Visayas
127		1	800	null
128		null	1500	null
129		null	7500	Mindanao
130		3	2200	null
131		1	45000	null
132		null	7500	Mindanao
133				

null

Query Settings

PROPERTIES

Name: MessySales (2)

All Properties

APPLIED STEPS

- Source
- Navigation
- Promoted Headers
- Changed Type

7 COLUMNS, 999 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED ON FRIDAY

Power Query – Data Cleaning Part 3 | Step 2. Apply appropriate action: Replace with 0 → Transform → Replace Values

Messy_Data_Power_BI

Home | Transform | Add Column | View | Tools | Help

Close & Apply | New Source | Recent Sources | Enter Data | Data source settings | Manage Parameters | Export query results | Refresh Preview | Properties | Advanced Editor | Manage | Choose Columns | Remove Columns | Keep Rows | Remove Rows | Split Column

Close | New Query | Data Sources | Parameters | Output Data | Query | Manage Columns | Reduce Rows | Sort

Queries [1]

MessySales

```
= Table.ReplaceValue(#"Removed Duplicates",null,"0",
```

	derID	OrderDate	CustomerName	Product
1	46	2025-01-19	Mia Tan	Monitor
2	23	2025-02-05	Paul Yu	Keyboard
3	79	08-02-2025	Pedro Reyes	Laptop
4	32	2025/02/27	Chris Go	Headset
5	16	13-01-2025	Ana Lopez	Webcam
6	13	2025-02-22	Bea Lim	Monitor
7	51	02/23/2025	Carlo Diaz	Headset
8	93	2025-02-08	MARIA SANTOS	Webcam
9	54	2025-01-24	Ella Yu	Mouse
10	04	2025-02-28	Ella Yu	Laptop
11	25	Feb 12 2025	Mia Tan	Monitor
12	60	2025-01-28	MARIA SANTOS	Webcam
13	50	Feb 13 2025	Chris Go	Headset
14	33	01-02-2025	Bea Lim	Mouse
15				

Query Settings

PROPERTIES

Name

MessySales

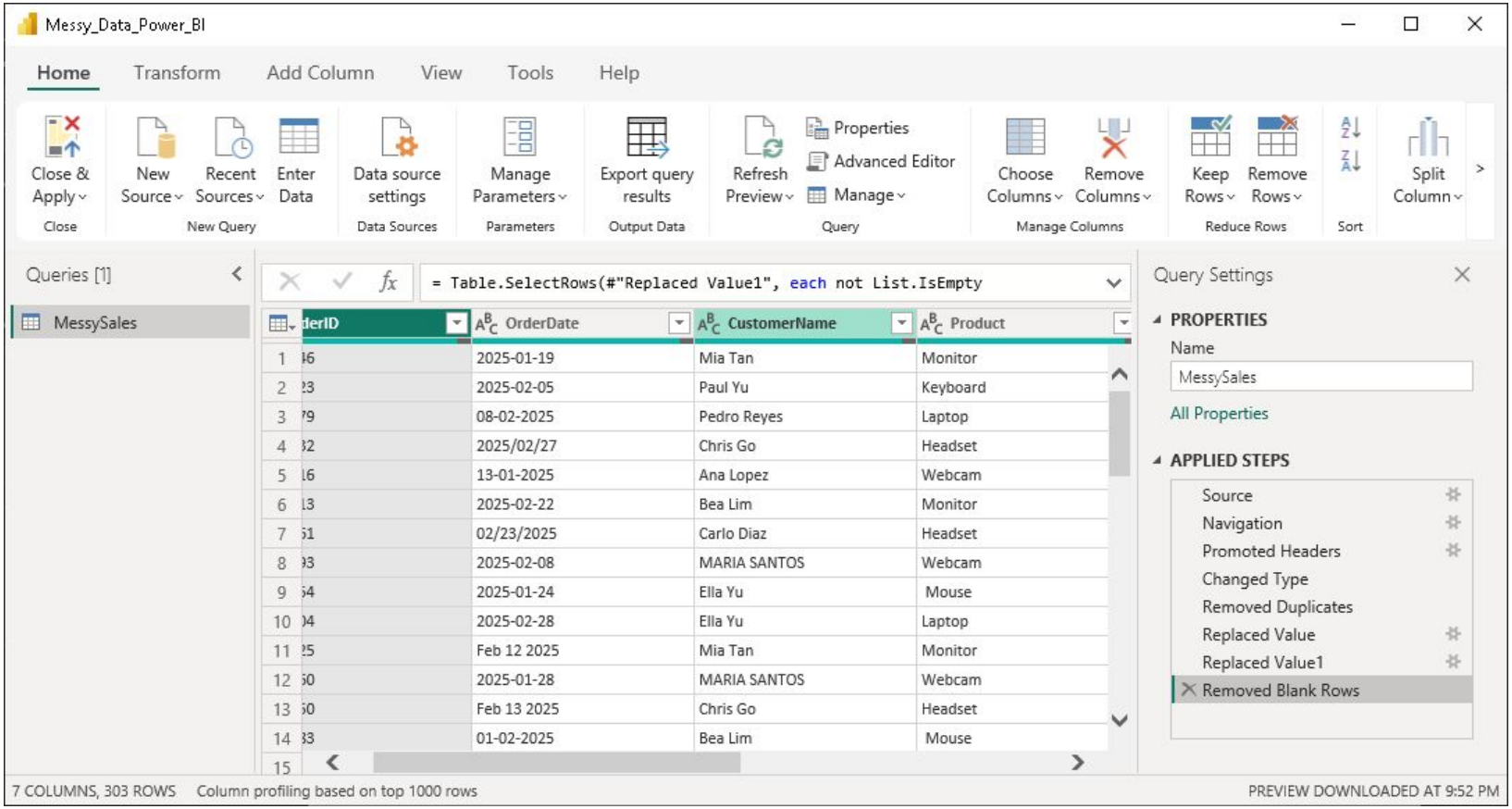
All Properties

APPLIED STEPS

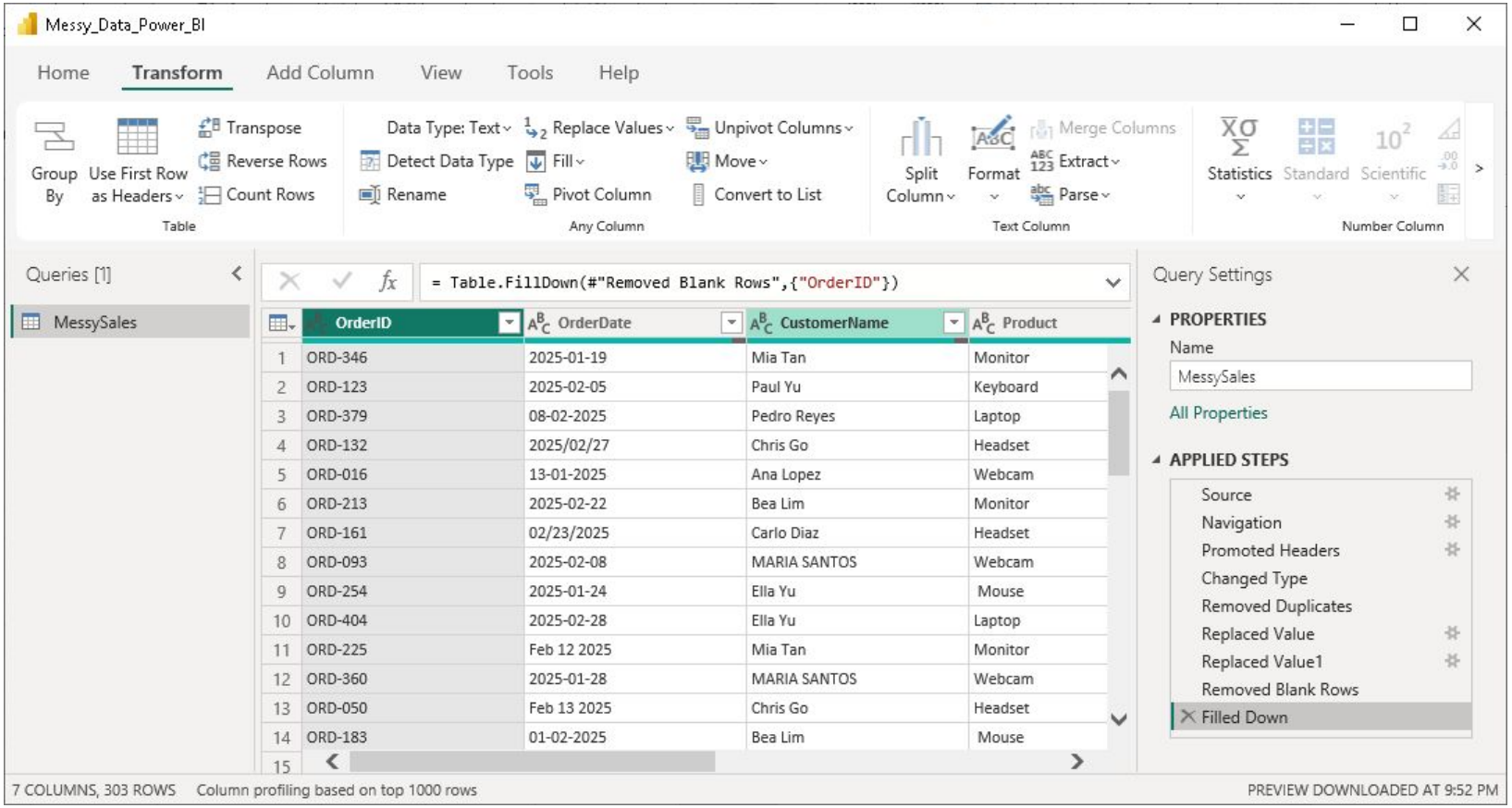
- Source
- Navigation
- Promoted Headers
- Changed Type
- Removed Duplicates
- Replaced Value
- Replaced Value1
- Removed Blank Rows

7 COLUMNS, 303 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 9:52 PM

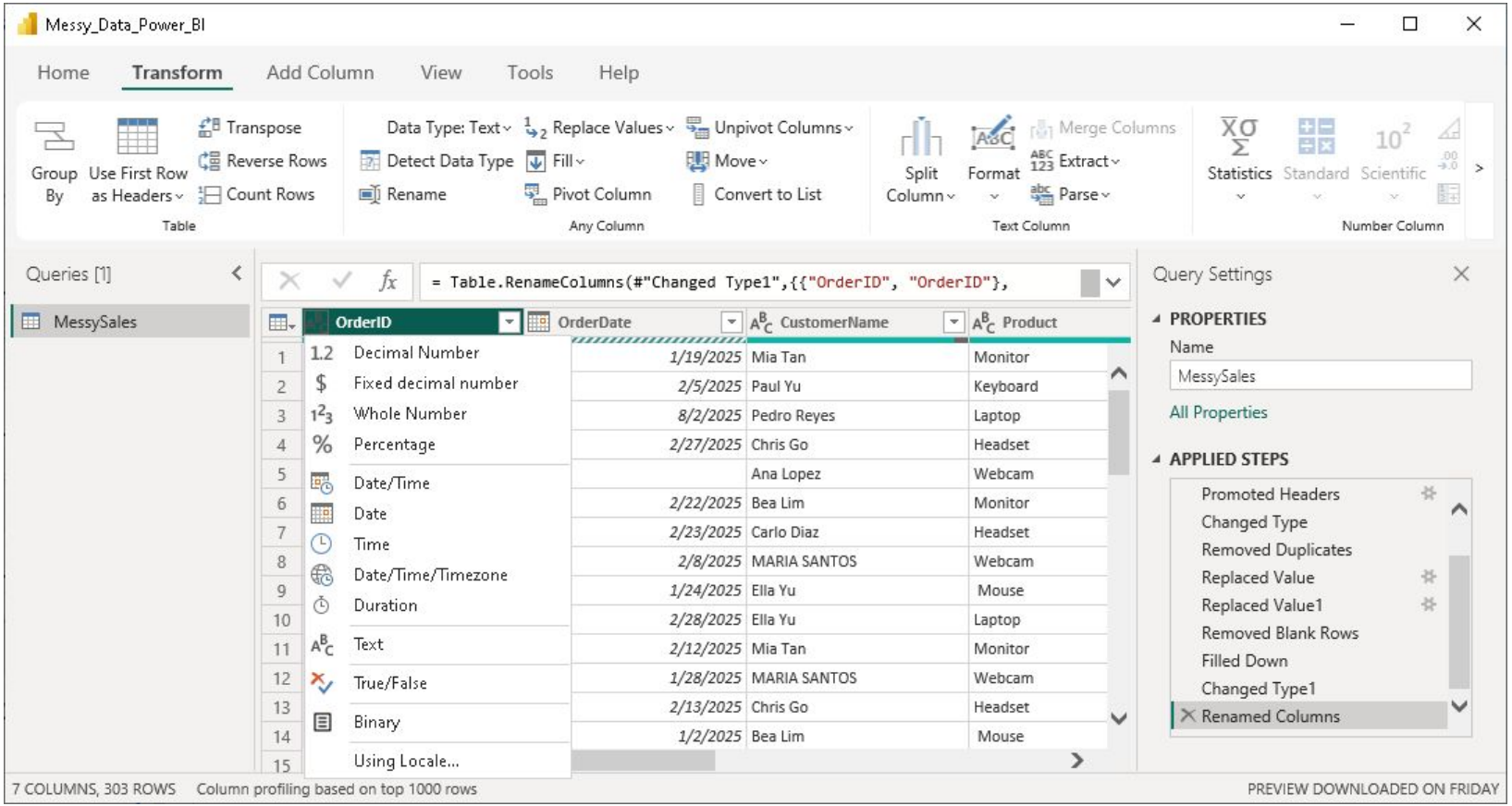
Power Query – Data Cleaning Part 3 | Step 2. Apply appropriate action: Remove rows → Remove Rows → Remove Blank Rows



Power Query – Data Cleaning Part 3 | Step 2. Apply appropriate action: Fill down → Transform → Fill → Down



Power Query – Data Cleaning Part 4 | Step 1. Check icons beside column names.



Power Query – Data Cleaning Part 4 | Step 2. Set correct types: Date → Date type

Messy_Data_Power_BI

Home | Transform | Add Column | View | Tools | Help

Group By | Use First Row as Headers | Transpose | Reverse Rows | Count Rows (Table)

Data Type: Date | Detect Data Type | Rename | Replace Values | Fill | Pivot Column | Unpivot Columns | Move | Convert to List (Any Column)

Split Column | Format | Merge Columns | Extract | Parse (Text Column)

Statistics | Standard | Scientific (Number Column)

Queries [1]

MessySales

```
= Table.TransformColumnTypes(#"Filled Down",{{"OrderDate", type date},
```

	OrderDate	CustomerName	Product
1	1/19/2025	Mia Tan	Monitor
2	2/5/2025	Paul Yu	Keyboard
3	8/2/2025	Pedro Reyes	Laptop
4	2/27/2025	Chris Go	Headset
5	Error	Ana Lopez	Webcam
6	2/22/2025	Bea Lim	Monitor
7	2/23/2025	Carlo Diaz	Headset
8	2/8/2025	MARIA SANTOS	Webcam
9	1/24/2025	Ella Yu	Mouse
10	2/28/2025	Ella Yu	Laptop
11	2/12/2025	Mia Tan	Monitor
12	1/28/2025	MARIA SANTOS	Webcam
13	2/13/2025	Chris Go	Headset
14	1/2/2025	Bea Lim	Mouse
15			

Query Settings

PROPERTIES

Name: MessySales

All Properties

APPLIED STEPS

- Navigation
- Promoted Headers
- Changed Type
- Removed Duplicates
- Replaced Value
- Replaced Value1
- Removed Blank Rows
- Filled Down
- Changed Type1

7 COLUMNS, 303 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 9:52 PM

Power Query – Data Cleaning Part 4 | Step 2. Set correct types: Sales → Decimal Number

Messy_Data_Power_BI

Home | Transform | Add Column | View | Tools | Help

Group By | Use First Row as Headers | Transpose | Reverse Rows | Count Rows (Table)

Data Type: Decimal Number | Detect Data Type | Rename | Replace Values | Fill | Pivot Column | Unpivot Columns | Move | Convert to List (Any Column)

Split Column | Format | Merge Columns | Extract | Parse (Text Column)

Statistics | Standard (Num)

Queries [1]

MessySales

= Table.TransformColumnTypes(#"Filled Down",{{"OrderDate", type

		Quantity	SalesAmount	Region
60		1	1200	0
61		0	3000	Mindanao
62		0	1500	Visayas
63		2	3000	0
64		2	3000	Mindanao
65		3	2200	0
66		1	Error	0
67		3	1500	Luzon
68		3	1200	0
69		0	800	0
70		0	2200	Mindanao
71		2	2200	0
72		1	1200	0
73		2	Error	Mindanao
74				

Query Settings

PROPERTIES

Name: MessySales

All Properties

APPLIED STEPS

- Navigation
- Promoted Headers
- Changed Type
- Removed Duplicates
- Replaced Value
- Replaced Value1
- Removed Blank Rows
- Filled Down
- Changed Type1

7 COLUMNS, 303 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 9:52 PM

Power Query – Data Cleaning Part 4 | Step 2. Set correct types: Quantity → Whole Number

Messy_Data_Power_BI

Home | Transform | Add Column | View | Tools | Help

Group By | Use First Row as Headers | Transpose | Reverse Rows | Count Rows | Table

Data Type: Whole Number | Detect Data Type | Rename | Replace Values | Fill | Pivot Column | Unpivot Columns | Move | Convert to List | Any Column

Split Column | Format | Merge Columns | Extract | Parse | Text Column

Statistics | Standard | Sci | Numbe

Queries [1]

MessySales

```
= Table.TransformColumnTypes(#"Filled Down",{{"OrderDate", type date},
```

	Quantity	SalesAmount	Region
1	2	7500	Luzon
2	1	45000	0
3	2	3000	Luzon
4	1	7500	Mindanao
5	1	7500	Mindanao
6	0	1500	Luzon
7	3	Error	0
8	2	800	Luzon
9	1	3000	Mindanao
10	3	2200	Visayas
11	3	3000	Mindanao
12	3	45000	Visayas
13	2	1200	0
14	3	7500	Visayas
15			

Query Settings

PROPERTIES

Name: MessySales

All Properties

APPLIED STEPS

- Navigation
- Promoted Headers
- Changed Type
- Removed Duplicates
- Replaced Value
- Replaced Value1
- Removed Blank Rows
- Filled Down
- Changed Type1

7 COLUMNS, 303 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 9:52 PM

Power Query – Data Cleaning Part 5 | Step 1. Double-click column names.

Messy_Data_Power_BI

Home | Transform | Add Column | View | Tools | Help

Table: Group By, Use First Row as Headers, Transpose, Reverse Rows, Count Rows

Any Column: Data Type: Text, Detect Data Type, Rename, Replace Values, Fill, Pivot Column, Unpivot Columns, Move, Convert to List

Text Column: Split Column, Format, Merge Columns, Extract, Parse

Number Column: Statistics, Standard, Scientific

Queries [1]

MessySales

```
= Table.TransformColumnTypes(#"Renamed Columns",{{"OrderID", type text}
    })
```

	OrderID	OrderDate	CustomerName	Product
70	ORD-381	1/10/2025	Bea Lim	Mouse
71	ORD-362	1/4/2025	Tom Lee	Headset
72	ORD-319	11/2/2025	Sarah Ong	Keyboard
73	ORD-212	2/7/2025	Pedro Reyes	Monitor
74	ORD-297	1/28/2025	Juan dela cruz	Mouse
75	ORD-100	1/31/2025	MARIA SANTOS	Laptop
76	ORD-157	1/30/2025	Ella Yu	Keyboard
77	ORD-178	2/9/2025	Tom Lee	Mouse
78	ORD-384	2/14/2025	Bea Lim	Keyboard
79	ORD-030	Error	MARIA SANTOS	Keyboard
80				

Query Settings

PROPERTIES

Name: MessySales

All Properties

APPLIED STEPS

- Changed Type
- Removed Duplicates
- Replaced Value
- Replaced Value1
- Removed Blank Rows
- Filled Down
- Changed Type1
- Renamed Columns
- Changed Type2

7 COLUMNS, 303 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED ON FRIDAY

Power Query – Data Cleaning Part 5 | Step 2. Use standard naming: order_id

Messy_Data_Power_BI

Home | Transform | Add Column | View | Tools | Help

Group By | Use First Row as Headers | Transpose | Reverse Rows | Count Rows (Table)

Data Type: Text | Detect Data Type | Rename | Replace Values | Fill | Pivot Column | Unpivot Columns | Move | Convert to List (Any Column)

Split Column | Format | Merge Columns | Extract | Parse (Text Column)

Statistics | Standard | Scientific (Number Column)

Queries [1]

MessySales

```
= Table.RenameColumns(#"Changed Type2",{{"OrderID", "order_id"},
    {"OrderDate", "order_date"}, {"CustomerName", "customer_name"}})
```

	order_id	order_date	customer_name	Product
1	ORD-346	1/19/2025	Mia Tan	Monitor
2	ORD-123	2/5/2025	Paul Yu	Keyboard
3	ORD-379	8/2/2025	Pedro Reyes	Laptop
4	ORD-132	2/27/2025	Chris Go	Headset
5	ORD-016	Error	Ana Lopez	Webcam
6	ORD-213	2/22/2025	Bea Lim	Monitor
7	ORD-161	2/23/2025	Carlo Diaz	Headset
8	ORD-093	2/8/2025	MARIA SANTOS	Webcam
9	ORD-254	1/24/2025	Ella Yu	Mouse
10	ORD-404	2/28/2025	Ella Yu	Laptop
11				

Query Settings

PROPERTIES

Name: MessySales

All Properties

APPLIED STEPS

- Removed Duplicates
- Replaced Value
- Replaced Value1
- Removed Blank Rows
- Filled Down
- Changed Type1
- Renamed Columns
- Changed Type2
- Renamed Columns1

7 COLUMNS, 303 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED ON FRIDAY

Power Query – Data Cleaning Part 5 | Step 2. Use standard naming: order_date

Messy_Data_Power_BI

Home | Transform | Add Column | View | Tools | Help

Group By | Use First Row as Headers | Transpose | Reverse Rows | Count Rows (Table)

Data Type: Date | Detect Data Type | Rename | Replace Values | Fill | Pivot Column | Unpivot Columns | Move | Convert to List (Any Column)

Split Column | Format | Merge Columns | Extract | Parse (Text Column)

Statistics | Standard | Scientific (Number Column)

Queries [1]

MessySales

```
= Table.RenameColumns(#"Changed Type1",{{"OrderID", "order_id"},
```

	order_id	order_date	customer_name	Product
1	ORD-346	1/19/2025	Mia Tan	Monitor
2	ORD-123	2/5/2025	Paul Yu	Keyboard
3	ORD-379	8/2/2025	Pedro Reyes	Laptop
4	ORD-132	2/27/2025	Chris Go	Headset
5	ORD-016	Error	Ana Lopez	Webcam
6	ORD-213	2/22/2025	Bea Lim	Monitor
7	ORD-161	2/23/2025	Carlo Diaz	Headset
8	ORD-093	2/8/2025	MARIA SANTOS	Webcam
9	ORD-254	1/24/2025	Ella Yu	Mouse
10	ORD-404	2/28/2025	Ella Yu	Laptop
11	ORD-225	2/12/2025	Mia Tan	Monitor
12	ORD-360	1/28/2025	MARIA SANTOS	Webcam
13	ORD-050	2/13/2025	Chris Go	Headset
14	ORD-183	1/2/2025	Bea Lim	Mouse
15				

Query Settings

PROPERTIES

Name: MessySales

All Properties

APPLIED STEPS

- Navigation
- Promoted Headers
- Changed Type
- Removed Duplicates
- Replaced Value
- Replaced Value1
- Removed Blank Rows
- Filled Down
- Changed Type1
- Renamed Columns

7 COLUMNS, 303 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED ON FRIDAY

Power Query – Data Cleaning Part 5 | Step 2. Use standard naming: customer_name

Messy_Data_Power_BI

Home | Transform | Add Column | View | Tools | Help

Table: Group By, Use First Row as Headers, Transpose, Reverse Rows, Count Rows

Any Column: Data Type: Text, Detect Data Type, Rename, Replace Values, Fill, Pivot Column, Unpivot Columns, Move, Convert to List

Text Column: Split Column, Format, Merge Columns, Extract, Parse

Number Column: Statistics, Standard, Scientific

Queries [1]

MessySales

```
= Table.RenameColumns(#"Changed Type1",{{"OrderID", "order_id"},
```

	order_id	order_date	customer_name	Product
1	ORD-346	1/19/2025	Mia Tan	Monitor
2	ORD-123	2/5/2025	Paul Yu	Keyboard
3	ORD-379	8/2/2025	Pedro Reyes	Laptop
4	ORD-132	2/27/2025	Chris Go	Headset
5	ORD-016	Error	Ana Lopez	Webcam
6	ORD-213	2/22/2025	Bea Lim	Monitor
7	ORD-161	2/23/2025	Carlo Diaz	Headset
8	ORD-093	2/8/2025	MARIA SANTOS	Webcam
9	ORD-254	1/24/2025	Ella Yu	Mouse
10	ORD-404	2/28/2025	Ella Yu	Laptop
11	ORD-225	2/12/2025	Mia Tan	Monitor
12	ORD-360	1/28/2025	MARIA SANTOS	Webcam
13	ORD-050	2/13/2025	Chris Go	Headset
14	ORD-183	1/2/2025	Bea Lim	Mouse
15				

Query Settings

PROPERTIES

Name: MessySales

All Properties

APPLIED STEPS

- Promoted Headers
- Changed Type
- Removed Duplicates
- Replaced Value
- Replaced Value1
- Removed Blank Rows
- Filled Down
- Changed Type1
- Renamed Columns

7 COLUMNS, 303 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED ON FRIDAY

Power Query – Data Cleaning Part 5 | Step 2. Use standard naming: total_sales

Messy_Data_Power_BI

Home | Transform | Add Column | View | Tools | Help

Group By | Use First Row as Headers | Transpose | Reverse Rows | Count Rows (Table)

Data Type: Decimal Number | Detect Data Type | Rename | Replace Values | Fill | Pivot Column | Unpivot Columns | Move | Convert to List (Any Column)

Split Column | Format | Merge Columns | Extract | Parse (Text Column)

Statistics | Standard (Num)

Queries [1]

MessySales

```
= Table.RenameColumns(#"Changed Type1",{{"OrderID", "order_id"},
```

	t	Quantity	total_sales	Region
1		2	7500	Luzon
2		1	45000	0
3		2	3000	Luzon
4		1	7500	Mindanao
5		1	7500	Mindanao
6		0	1500	Luzon
7		3	Error	0
8		2	800	Luzon
9		1	3000	Mindanao
10		3	2200	Visayas
11		3	3000	Mindanao
12		3	45000	Visayas
13		2	1200	0
14		3	7500	Visayas
15				

Query Settings

PROPERTIES

Name: MessySales

All Properties

APPLIED STEPS

- Promoted Headers
- Changed Type
- Removed Duplicates
- Replaced Value
- Replaced Value1
- Removed Blank Rows
- Filled Down
- Changed Type1
- Renamed Columns

7 COLUMNS, 303 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED ON FRIDAY

Guide Questions

1. Why is removing duplicates important in data analysis?

Removing duplicates is critical for several reasons:

Accuracy of aggregations: Duplicate rows inflate counts, sums, and averages. For example, if a $500 sale appears twice, your revenue reports will be incorrect by $500.

Reliable insights: Decision-makers rely on trustworthy metrics. Duplicates create artificial trends and skew statistical analysis.

Performance optimization: Fewer rows mean faster query processing, smaller file sizes, and improved report refresh times in Power BI.

Data integrity: Duplicates often indicate upstream data collection issues that need addressing at the source.

Guide Questions

2. What problems can missing values cause in reports?

Missing values (nulls/blanks) create significant issues:

Broken visualizations: Charts may display gaps, errors, or misleading trends when values are absent.

Calculation failures: DAX measures using SUM, AVERAGE, or DIVIDE can return blanks or errors when encountering nulls.

Incorrect filtering: Blank values appear as "(Blank)" in slicers, confusing users and cluttering the interface.

Distorted analysis: Excluding rows with missing values can introduce bias; including them can skew results.

Relationship issues: Missing keys prevent proper table relationships in your data model.

Guide Questions

3. How does correct data typing affect calculations?

Data types fundamentally determine how Power BI processes your data:

Text

- Prevents mathematical operations

  Example: "100" + "200" = "100200" (concatenation), not 300

Whole Number/Decimal

- Enables arithmetic and statistical functions
- Ensures proper numerical sorting

Guide Questions

3. How does correct data typing affect calculations?

Data types fundamentally determine how Power BI processes your data:

Date/DateTime

- Allows time intelligence functions (YTD, MoM, YoY)
- Enables proper chronological sorting and date hierarchies

Boolean

- Enables true/false logic for conditional calculations
- Supports IF statements and logical operations

Guide Questions

3. How does correct data typing affect calculations?

Data types fundamentally determine how Power BI processes your data:

Currency

- Applies fixed decimal precision for financial accuracy.
- Prevents floating-point rounding errors in monetary calculations.

Common pitfalls:

- Imported numbers stored as text cannot be summed until converted.
- Dates stored as text won't work with DAX time intelligence functions like SAMEPERIODLASTYEAR().
- Mixed types in a column cause errors or unexpected behavior.

Guide Questions

4. What naming conventions improve data readability?

Consistent naming transforms cryptic databases into business-friendly models:

Table Names:

- Use PascalCase: SalesOrders, CustomerDetails (not sales_orders or tblSales).
- Avoid prefixes like "tbl", "dim", "fact" — users don't need technical schema knowledge.
- Use plural for fact tables (Sales), singular for dimensions (Customer).

Column Names:

- Descriptive and self-documenting: OrderDate instead of Dt, CustomerName instead of CustNm.
- No spaces (enables cleaner DAX): ProductCategory not Product Category.
- Standardize abbreviations: consistently use ID, Num, Amt (or spell out fully).

Guide Questions

4. What naming conventions improve data readability?

Consistent naming transforms cryptic databases into business-friendly models:

Measure Names:

- Include calculation type: Total Sales, Average Order Value, YTD Revenue
- Use units where helpful: Sales Amount USD, Quantity Units*
- Group related measures with prefixes: Sales Total, Sales YoY %, Sales per Customer

Enhancement Activities

- **Create a data quality checklist for future datasets**

Pre- Import Verification:

- [ ] Check for duplicate header rows — Ensure column names don't appear as data rows
- [ ] Verify unique identifiers — Confirm OrderID/primary keys are truly unique (found 198 duplicates in 500 rows)
- [ ] Check for missing values — Identify blank cells in critical fields (found 26.6% missing Quantity, 40.2% missing Region)

Data Type Validation:

- [ ] Numeric columns as numbers — Verify SalesAmount isn't stored as text with quotes (found "7500" with quotation marks)
- [ ] Date columns as dates — Ensure OrderDate is datetime type, not text
- [ ] Check for whitespace — Trim leading/trailing spaces in text fields (found " Mouse " with spaces in Product column)

Enhancement Activities

- **Create a data quality checklist for future datasets**

Standardization Checks:

- [ ] Consistent capitalization — Standardize names (found "MARIA SANTOS" in ALL CAPS alongside normal case)
- [ ] Consistent naming conventions — Verify consistent spelling and spacing (e.g., "Juan dela cruz" vs "Juan Dela Cruz")
- [ ] Categorical consistency — Check Region values are standardized (Luzon, Visayas, Mindanao)

Business Logic Validation:

- [ ] Reasonable value ranges — Check SalesAmount and Quantity are positive and within expected ranges
- [ ] Date range validity — Ensure dates fall within expected period (Jan-Mar 2025)

Enhancement Activities

- Compare dataset size before and after cleaning

Power Query Data Cleaning: Messy_Sales_Data.xlsx
(Lab Activities Results)

Metric	Before Cleaning	After Cleaning	Improvement
Row Count	500	302	39.6% reduction
Duplicate OrderIDs	198	0	100% eliminated
Missing Quantity (After Dedup)	82	0 (filled with 0)	100% filled
Missing Region (After Dedup)	117	117 (still missing)	0% (not handled)
SalesAmount as Text	49	0 (converted)	100% converted
Date Format Issues	302 (multiple formats)	302 (standardized)	100% standardized

Enhancement Activities

- ## Research: What is ETL and how does Power Query support it?

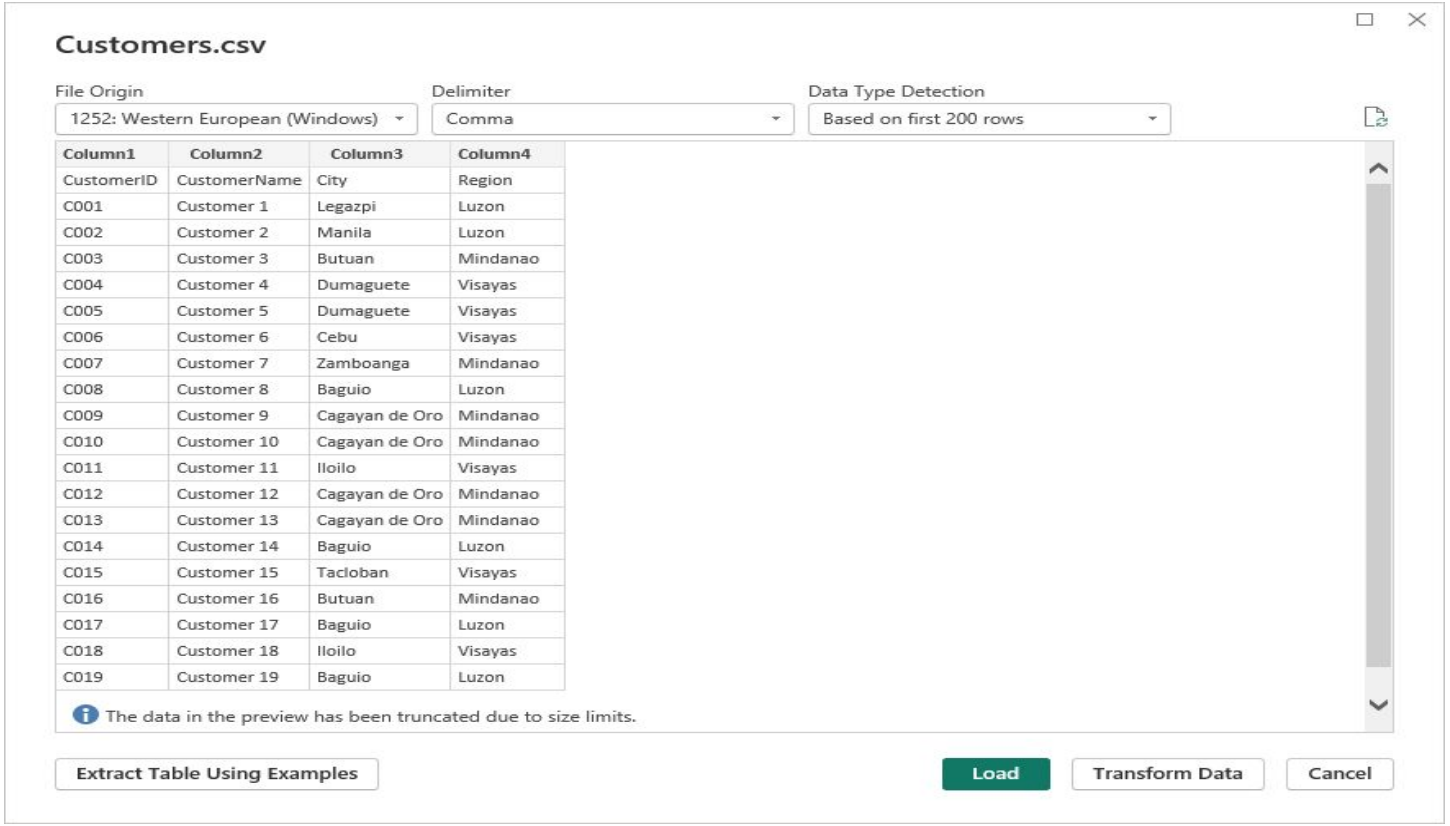
ETL stands for Extract, Transform, Load — the three fundamental steps of data integration. ETL is the backbone of business intelligence. Power Query is Power BI's ETL engine that allowed you to:

Extract 500 messy rows from Excel.

Transform them through 5 cleaning steps (removing 198 duplicates, filling 82 missing quantities, converting 49 text values, standardizing dates, renaming columns).

Load 302 clean rows into your data model ready for visualization.

Merge and Append Queries Part 1 | Step 1. Import both CSV files using Get Data.



Column1	Column2	Column3	Column4
CustomerID	CustomerName	City	Region
C001	Customer 1	Legazpi	Luzon
C002	Customer 2	Manila	Luzon
C003	Customer 3	Butuan	Mindanao
C004	Customer 4	Dumaguete	Visayas
C005	Customer 5	Dumaguete	Visayas
C006	Customer 6	Cebu	Visayas
C007	Customer 7	Zamboanga	Mindanao
C008	Customer 8	Baguio	Luzon
C009	Customer 9	Cagayan de Oro	Mindanao
C010	Customer 10	Cagayan de Oro	Mindanao
C011	Customer 11	Iloilo	Visayas
C012	Customer 12	Cagayan de Oro	Mindanao
C013	Customer 13	Cagayan de Oro	Mindanao
C014	Customer 14	Baguio	Luzon
C015	Customer 15	Tacloban	Visayas
C016	Customer 16	Butuan	Mindanao
C017	Customer 17	Baguio	Luzon
C018	Customer 18	Iloilo	Visayas
C019	Customer 19	Baguio	Luzon

Merge and Append Queries Part 1 | Step 2. Open Power Query Editor.

Untitled - Power Query Editor

Home | Transform | Add Column | View | Tools | Help

Close & Apply | New Source | Recent Sources | Enter Data | Data source settings | Manage Parameters | Export query results | Refresh Preview | Properties | Advanced Editor | Manage | Choose Columns | Remove Columns | Keep Rows | Remove Rows | Split Column

Close | New Query | Data Sources | Parameters | Output Data | Query | Manage Columns | Reduce Rows | Sort

Queries [2]

- Customers
- Orders

= Table.TransformColumnTypes(Source,{{"Column1", type text},

	Column1	Column2	Column3	Column4
1	CustomerID	CustomerName	City	Region
2	C001	Customer 1	Legazpi	Luzon
3	C002	Customer 2	Manila	Luzon
4	C003	Customer 3	Butuan	Mindanao
5	C004	Customer 4	Dumaguete	Visayas
6	C005	Customer 5	Dumaguete	Visayas
7	C006	Customer 6	Cebu	Visayas
8	C007	Customer 7	Zamboanga	Mindanao
9	C008	Customer 8	Baguio	Luzon
10	C009	Customer 9	Cagayan de Oro	Mindanao
11	C010	Customer 10	Cagayan de Oro	Mindanao
12	C011	Customer 11	Iloilo	Visayas
13	C012	Customer 12	Cagayan de Oro	Mindanao
14	C013	Customer 13	Cagayan de Oro	Mindanao
15				

Query Settings

PROPERTIES

Name: Customers

All Properties

APPLIED STEPS

- Source
- Changed Type

4 COLUMNS, 201 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 6:07 PM

Merge and Append Queries Part 2 | Step 1. Select Orders table.

Untitled - Power Query Editor

Home | Transform | Add Column | View | Tools | Help

Close & Apply | New Source | Recent Sources | Enter Data | Data source settings | Manage Parameters | Export query results | Refresh Preview | Properties | Advanced Editor | Manage | Choose Columns | Remove Columns | Keep Rows | Remove Rows | Split Column

Close | New Query | Data Sources | Parameters | Output Data | Query | Manage Columns | Reduce Rows | Sort

Queries [2]

- Customers
- Orders

= Table.TransformColumnTypes(#"Promoted Headers",{{"OrderID", type

	OrderID	CustomerID	OrderDate	SalesAmount
1	O001	C009	1/4/2025	
2	O002	C156	2/12/2025	
3	O003	C137	3/11/2025	
4	O004	C127	1/1/2025	
5	O005	C105	3/3/2025	
6	O006	C054	3/29/2025	
7	O007	C091	2/7/2025	
8	O008	C016	1/17/2025	
9	O009	C027	3/29/2025	
10	O010	C094	2/19/2025	
11	O011	C094	2/22/2025	
12	O012	C146	3/1/2025	
13	O013	C131	3/7/2025	
14	O014	C170	2/12/2025	
15				

Query Settings

PROPERTIES

Name: Orders

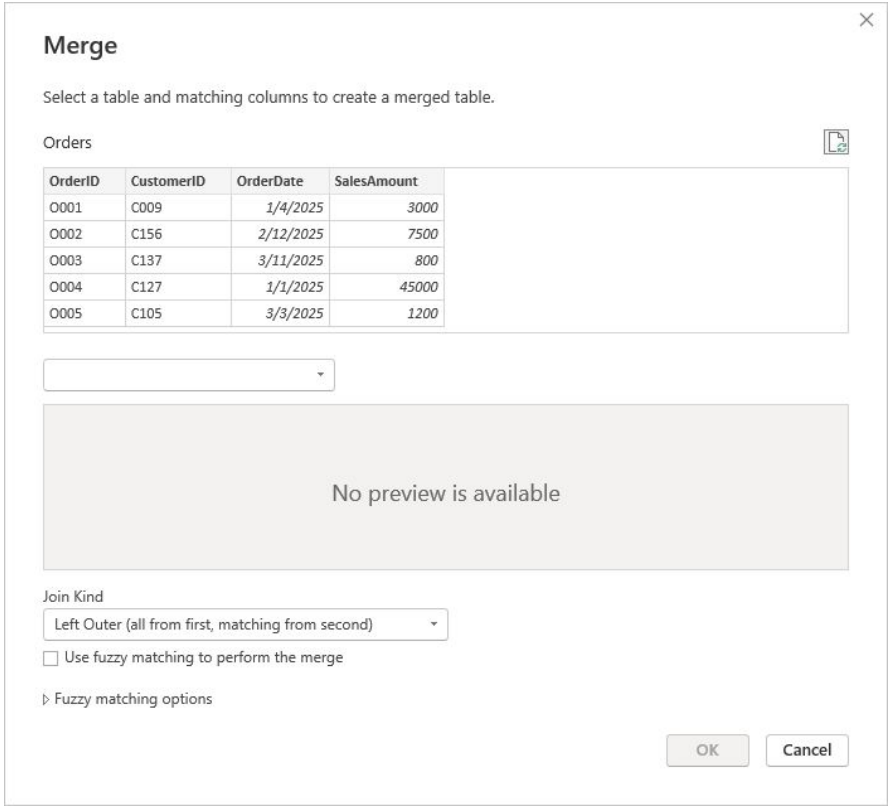
All Properties

APPLIED STEPS

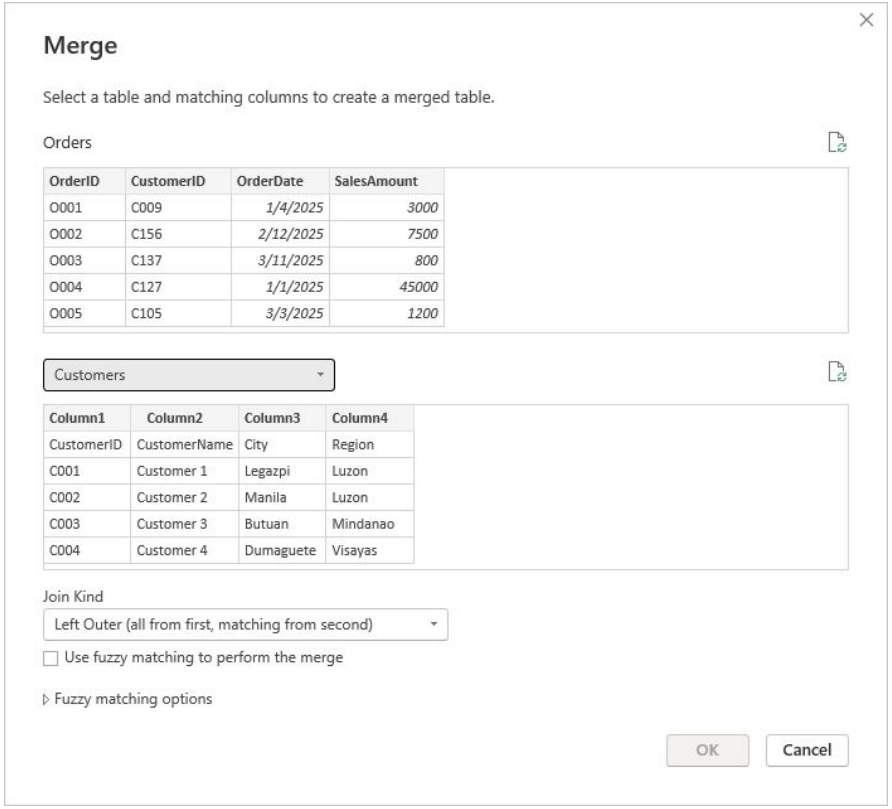
- Source
- Promoted Headers
- Changed Type

4 COLUMNS, 400 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 6:09 PM

Merge and Append Queries Part 2 | Step 2. Click Home → Merge Queries.



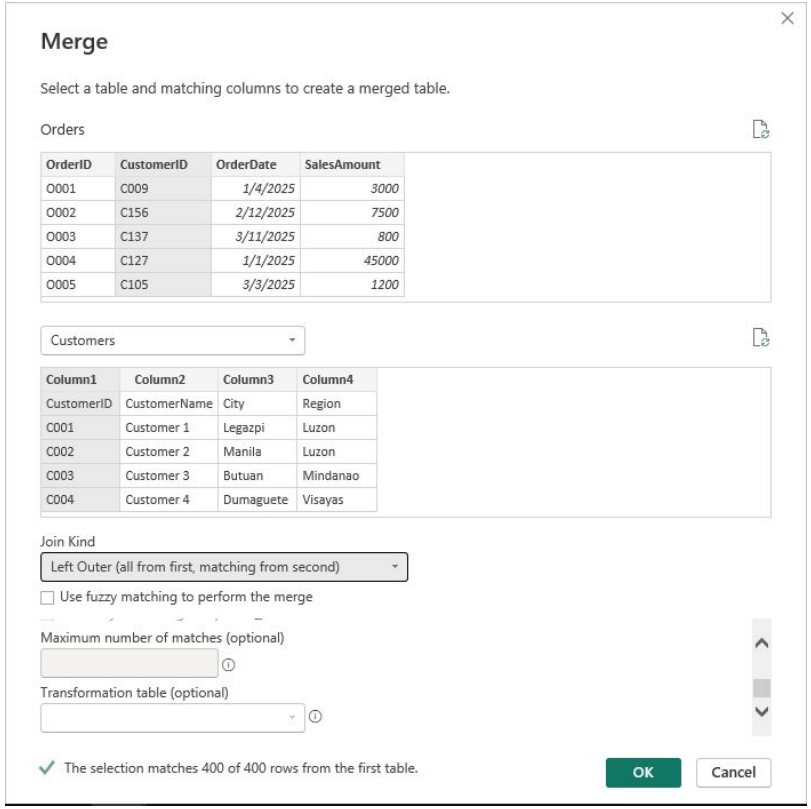
Merge and Append Queries Part 2 | Step 3. Select Customers table.



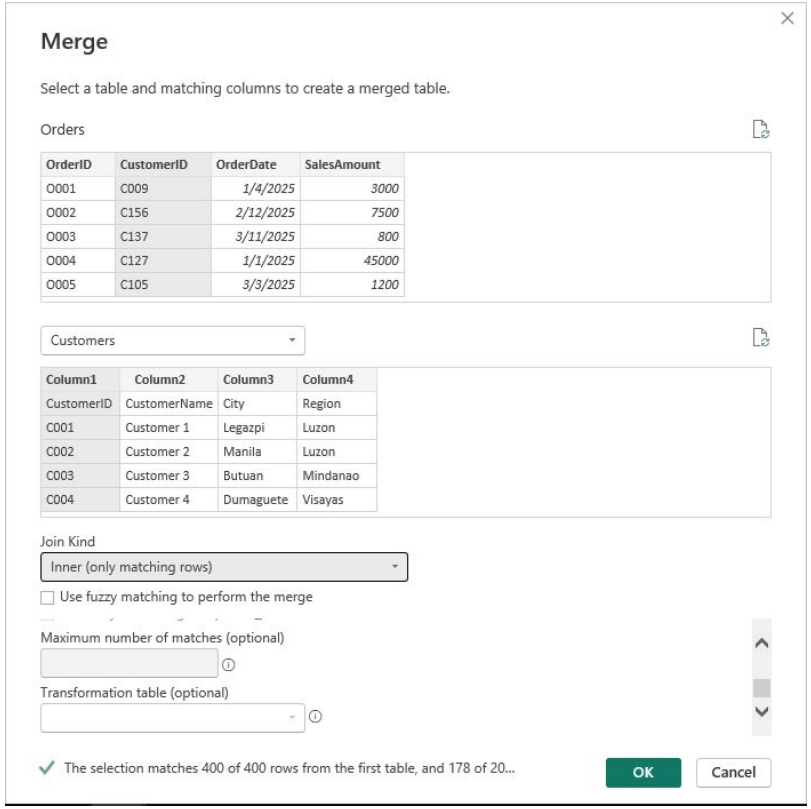
OrderID	CustomerID	OrderDate	SalesAmount
O001	C009	1/4/2025	3000
O002	C156	2/12/2025	7500
O003	C137	3/11/2025	800
O004	C127	1/1/2025	45000
O005	C105	3/3/2025	1200

Column1	Column2	Column3	Column4
CustomerID	CustomerName	City	Region
C001	Customer 1	Legazpi	Luzon
C002	Customer 2	Manila	Luzon
C003	Customer 3	Butuan	Mindanao
C004	Customer 4	Dumaguete	Visayas

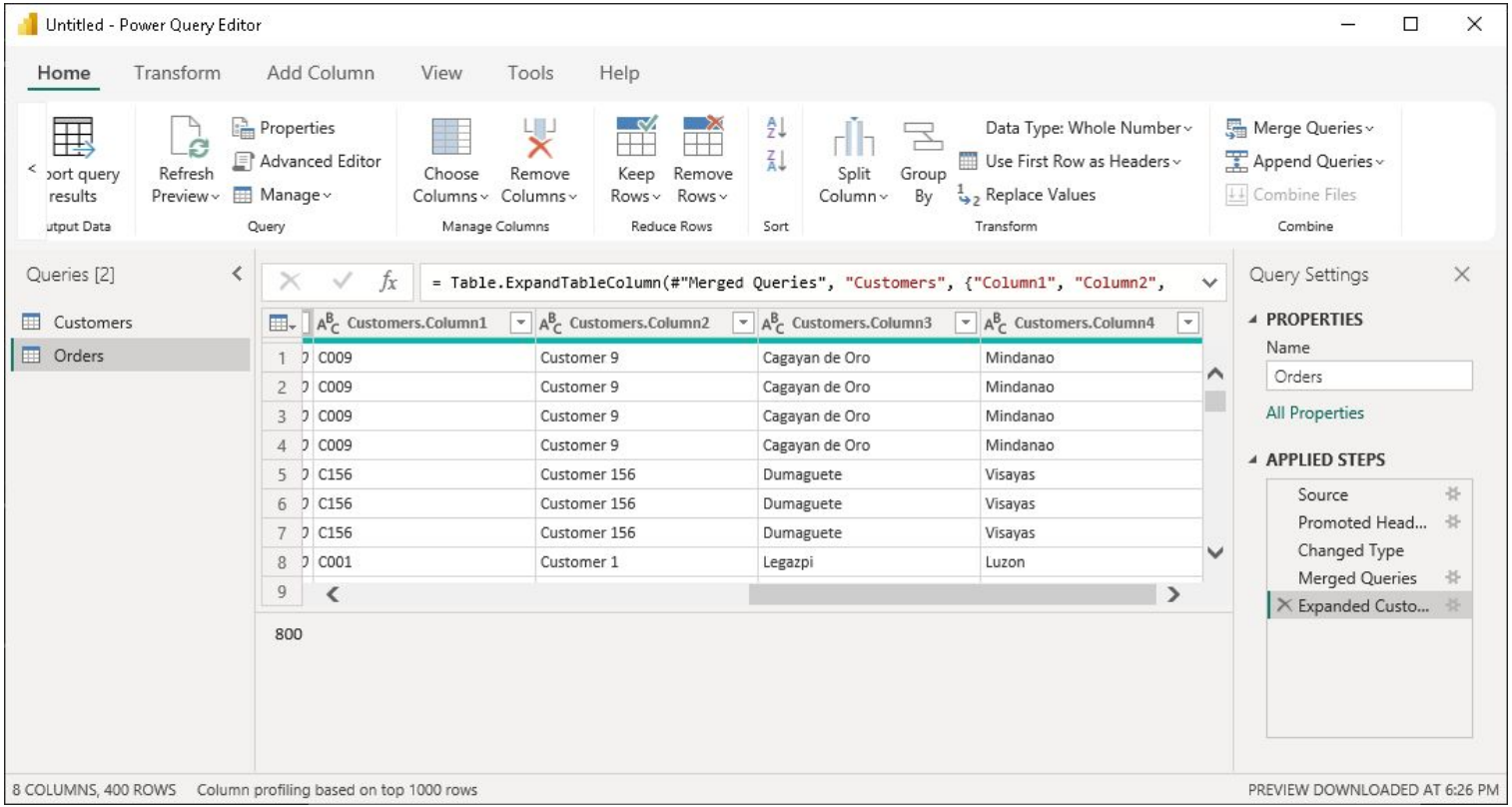
Merge and Append Queries Part 2 | Step 4. Match using CustomerID.



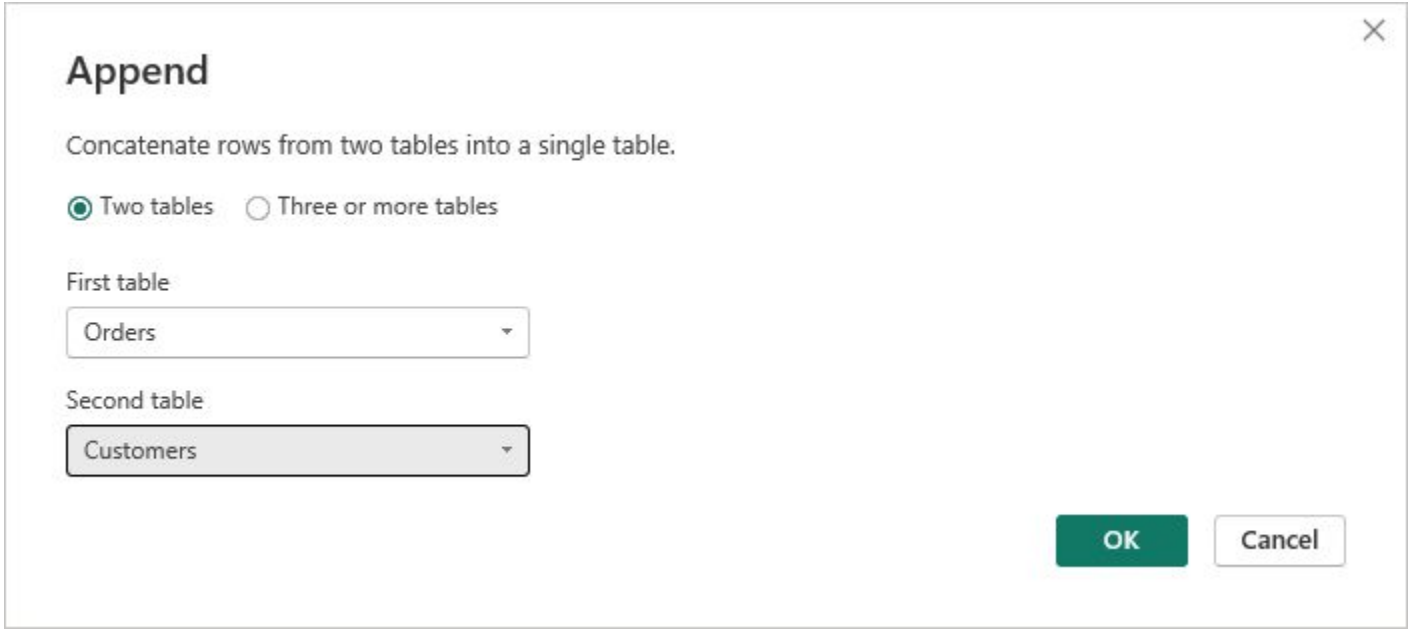
Merge and Append Queries Part 2 | Step 5. Choose Join Kind: Inner Join.



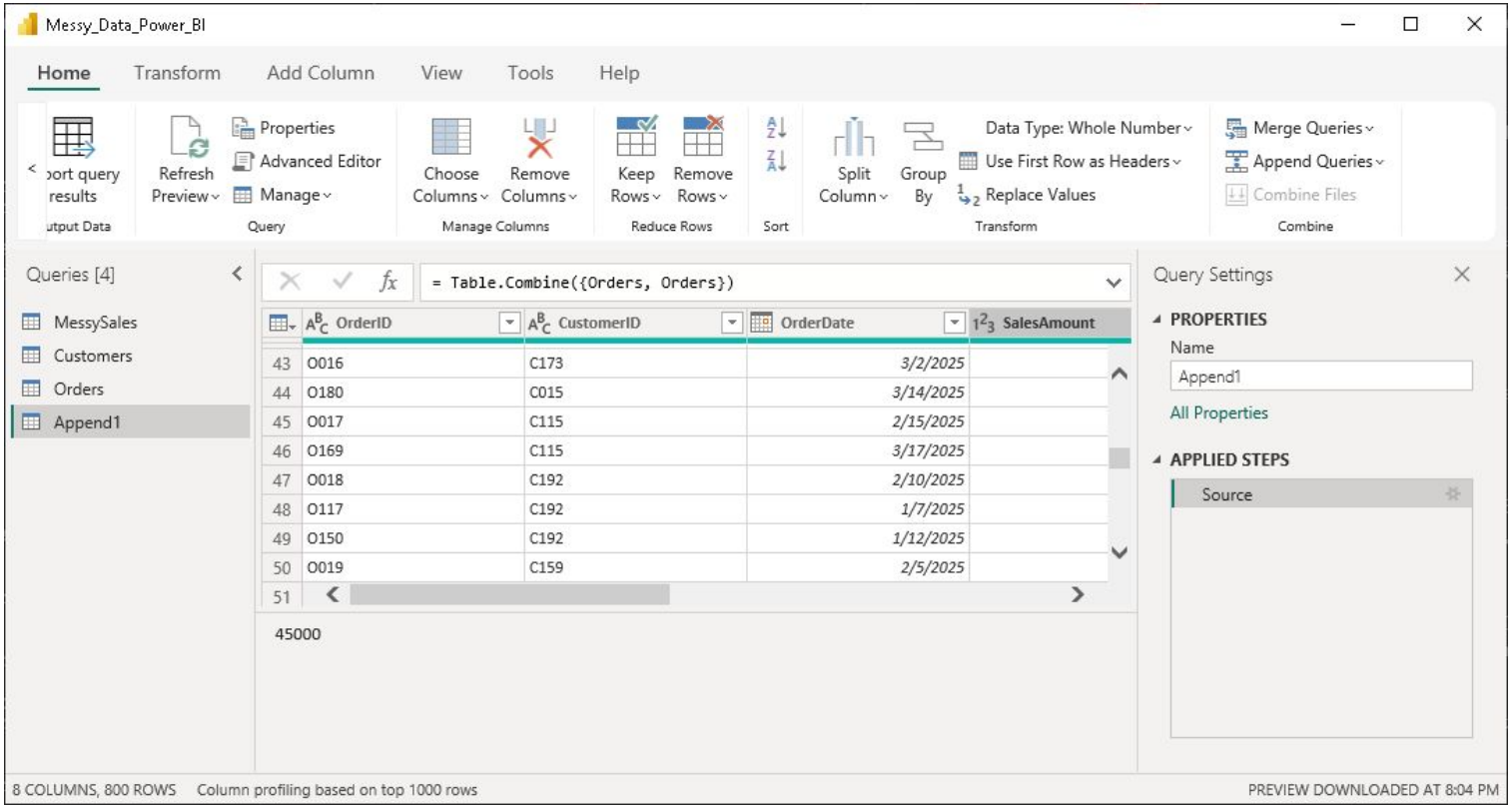
Merge and Append Queries Part 2 | Step 6. Expand merged table to include customer details.



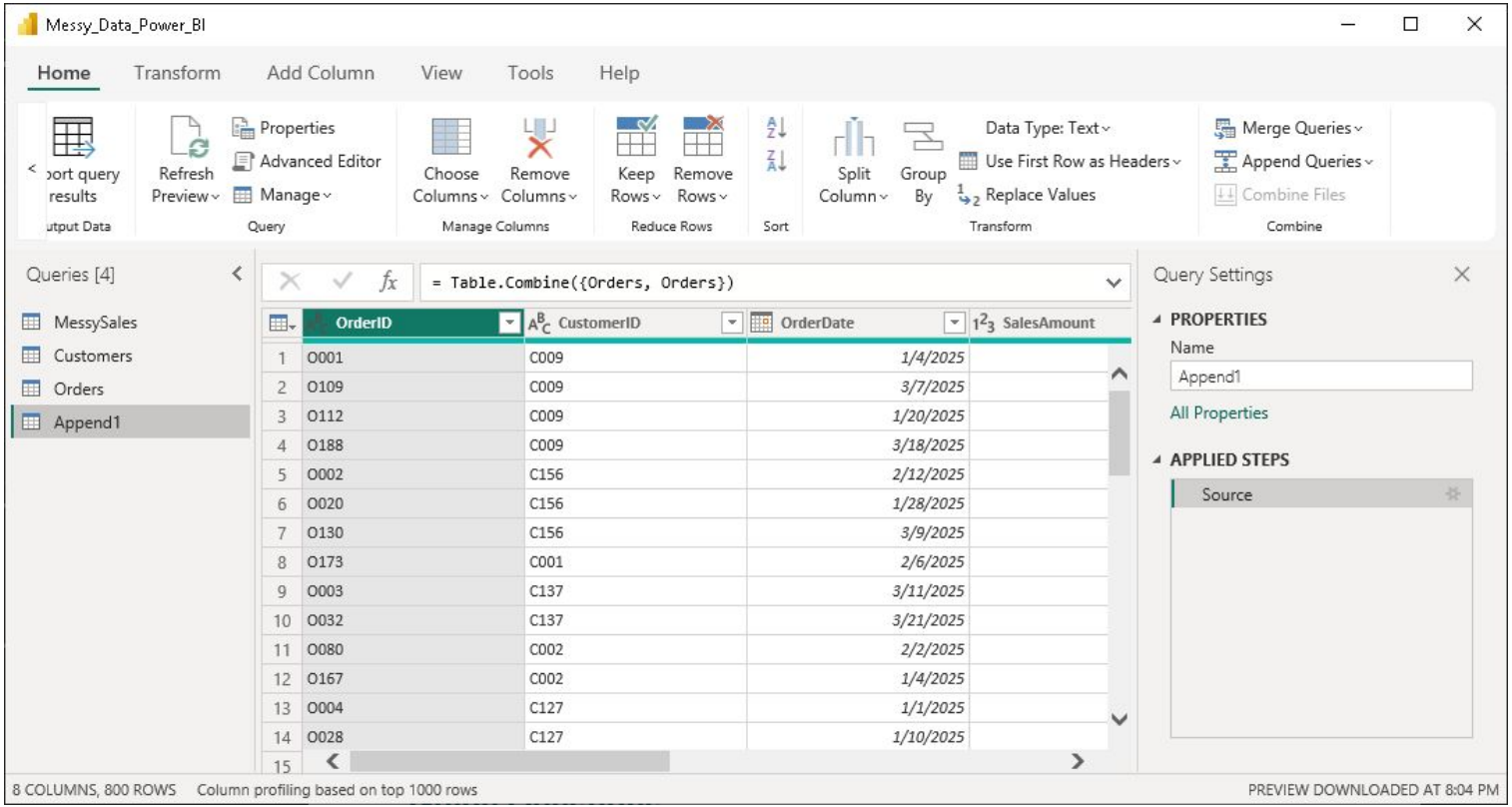
Merge and Append Queries Part 3 | Step 1. Click Append Queries → Append as New.



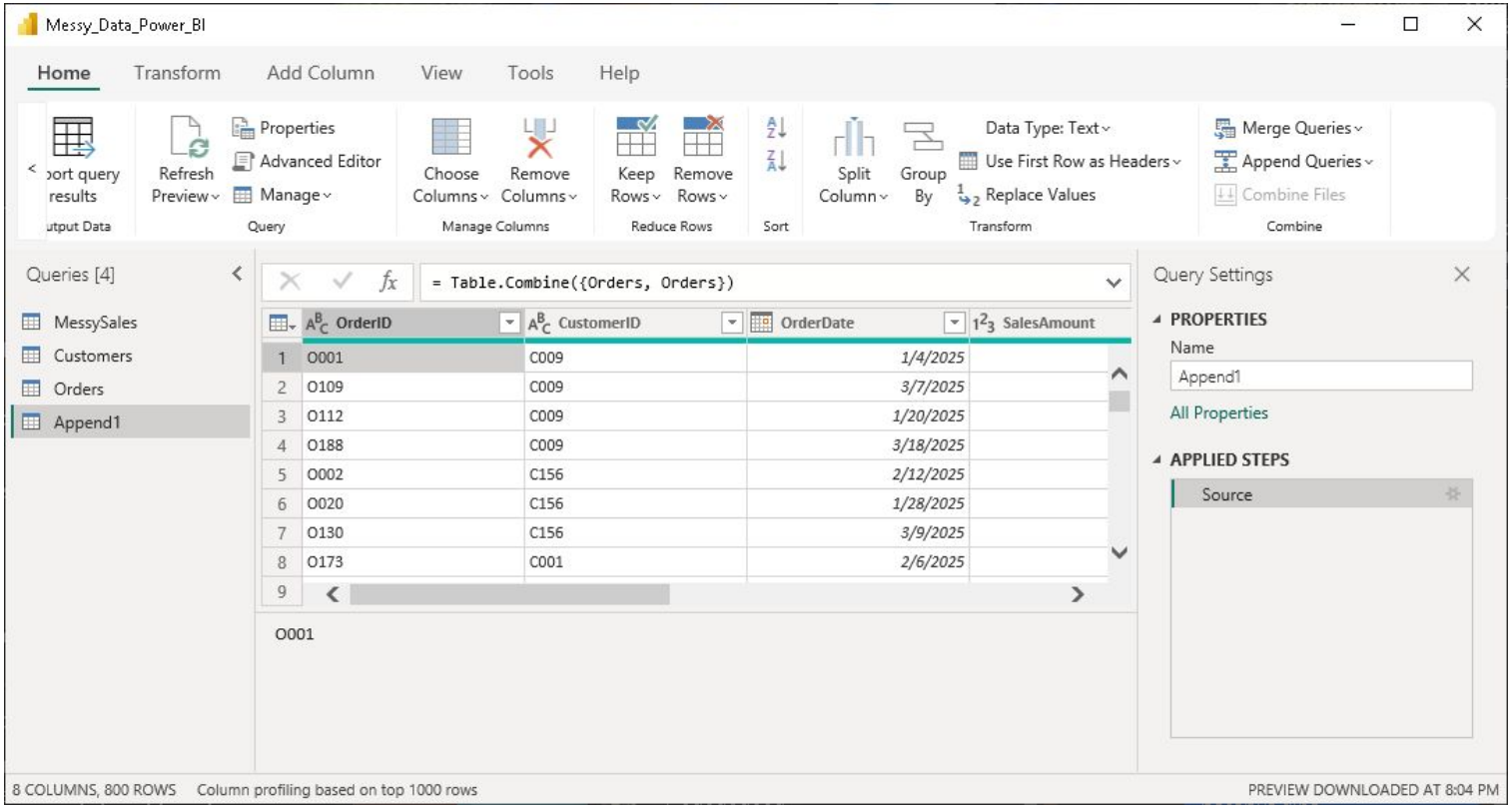
Merge and Append Queries Part 3 | Step 2. Select multiple monthly sales tables.



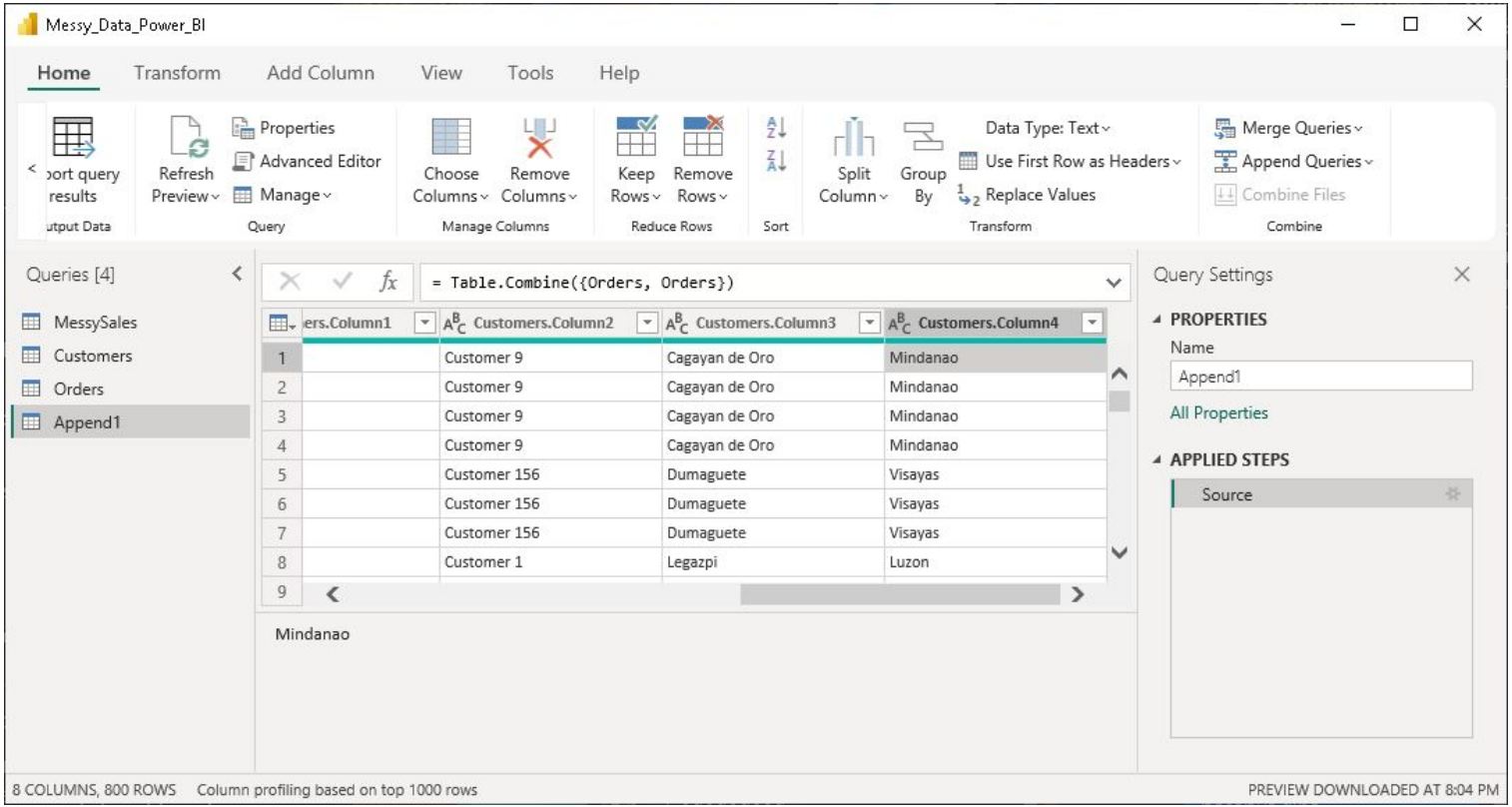
Merge and Append Queries Part 3 | Step 3. Confirm the combined table.



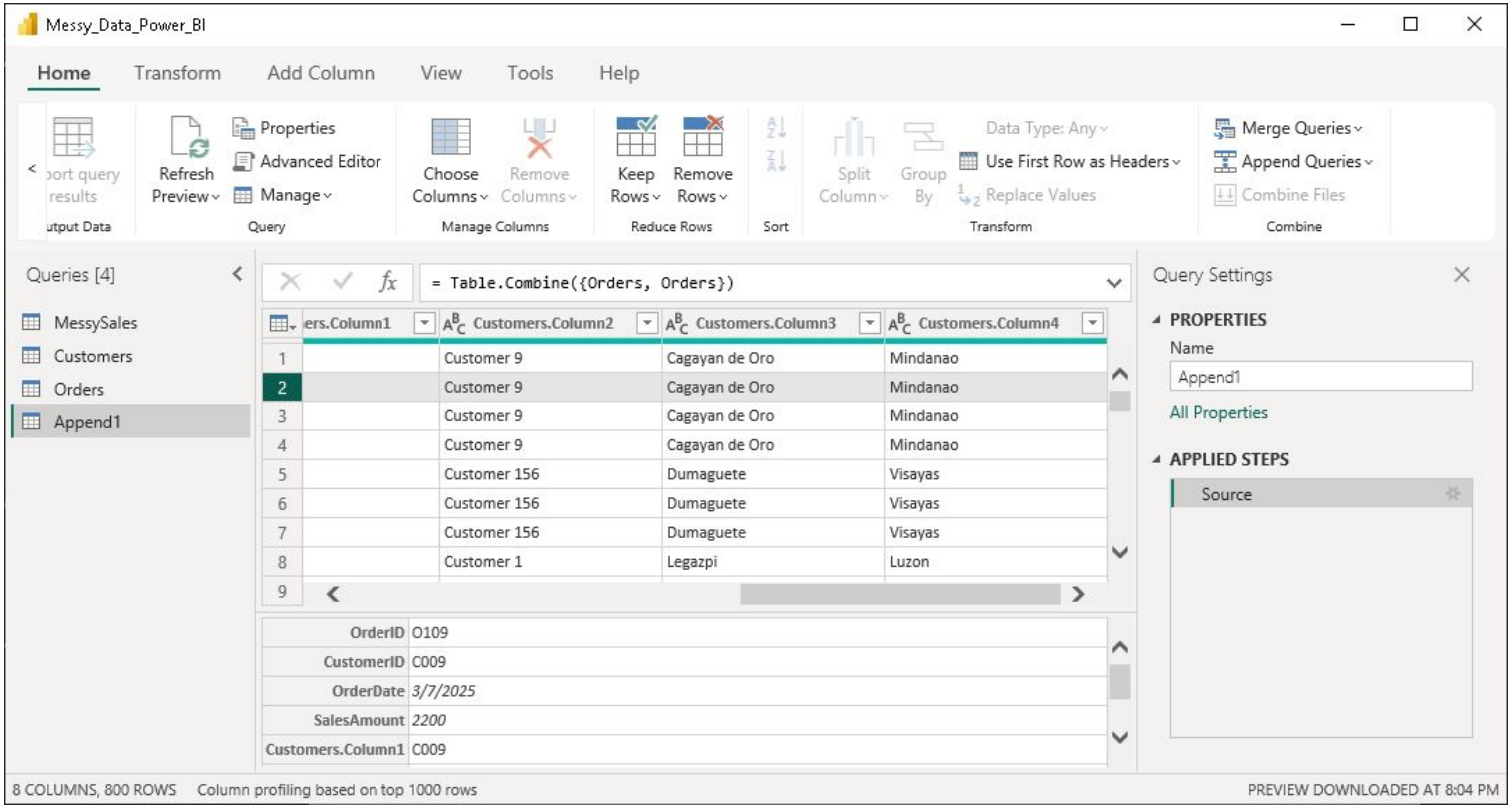
Merge and Append Queries Part 4 | Step 1. Check row counts.



Merge and Append Queries Part 4 | Step 2. Verify no unexpected null values.



Merge and Append Queries Part 4 | Step 3. Confirm matching customer details.



Guide Questions

1. What is the difference between Merge and Append?

Merge and Append are two different ways to combine data in Power Query. Merge combines tables horizontally by adding columns from one table to another based on a matching key column, similar to how VLOOKUP or SQL JOIN works. For example, if you have a Sales table and a Customers table, merging them would add customer details like name and city to each sales record. Append, on the other hand, combines tables vertically by stacking rows on top of each other, which works best when you have multiple tables with the same column structure, such as combining January sales data, February sales data, and March sales data into one master sales table. The key distinction is that Merge requires a common column to match records and results in a wider table with more columns, while Append requires identical column names and results in a taller table with more rows.

Guide Questions

2. Why use Inner Join instead of Left Join?

You should use Inner Join when you only want to keep records that have matching data in both tables, ensuring complete and consistent information for every row in your result. Inner Join is preferred over Left Join when you want to eliminate orphaned records that would have null values, reduce your dataset size by excluding incomplete records, or prevent calculation errors that occur when DAX formulas encounter blank cells. For instance, if you are merging sales orders with customer information and choose Inner Join, you will only see orders that have valid customers assigned, filtering out any test orders or data entry errors that lack proper customer details. This creates a cleaner, more reliable dataset for reporting and analysis, though you should ensure that excluding unmatched records does not remove important data that your business needs to track.

Guide Questions

3. What happens to unmatched records in an Inner Join?

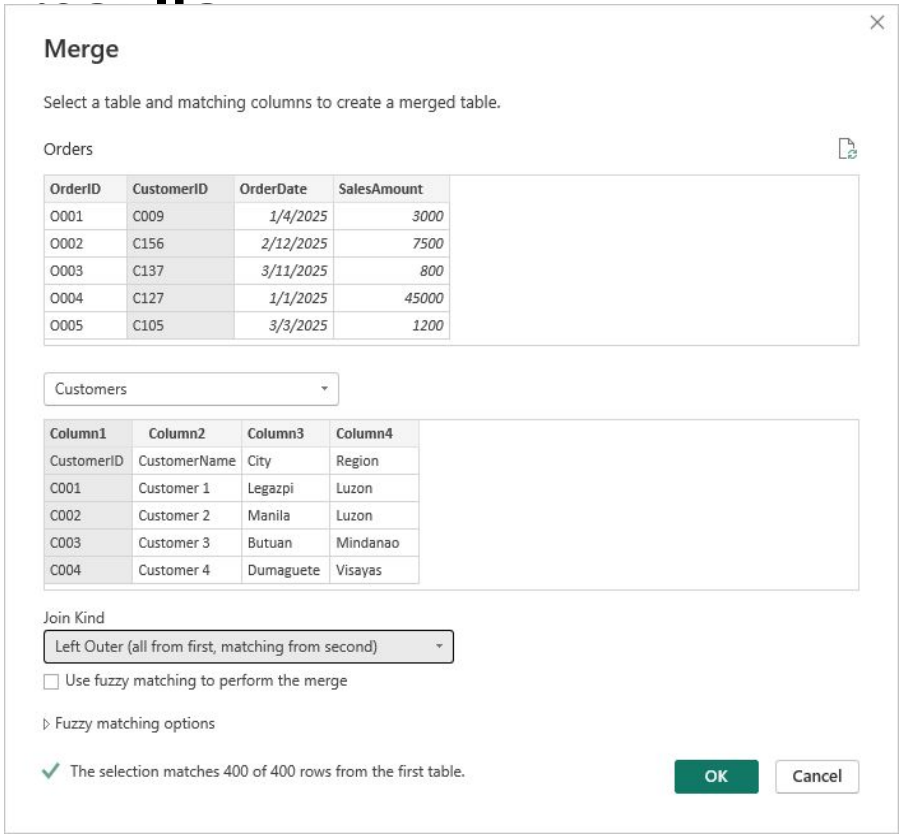
In an Inner Join, unmatched records are completely excluded from the final result set and do not appear in your merged query. When Power Query performs an Inner Join, it compares the key columns from both tables and only keeps rows where the key values exist in both the primary table and the lookup table. Any record from the first table that does not find a match in the second table is filtered out and discarded from the output, and likewise, any record from the second table that does not match the first table is also excluded. For example, if you have a sales order with a CustomerID that does not exist in your Customers table, that entire sales record will disappear from your merged result. These excluded records are not deleted from your original source data; they simply do not appear in this particular query, though you can create a separate Left Join query if you need to analyze or identify those unmatched records.

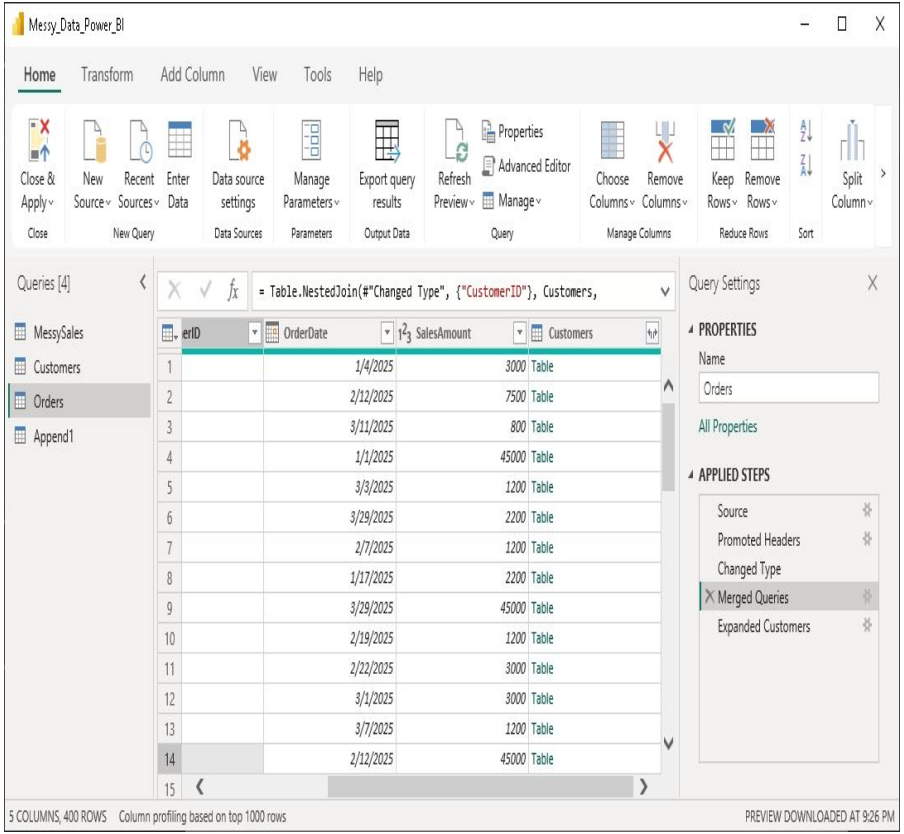
Guide Questions

4. When is Append more appropriate than Merge?

Append is more appropriate than Merge when you need to combine multiple tables that share the same structure but contain different sets of rows, such as combining monthly sales reports, regional data files, or historical records from different time periods. You should choose Append when your goal is to increase the volume of data by adding more observations rather than enriching existing records with additional attributes. Common scenarios include stacking daily transaction logs into a master database, combining quarterly financial reports into an annual summary, or consolidating sales data from different branches that all use the same table format. Append is the right choice when your tables have identical or nearly identical column names and data types, whereas Merge should be used when you need to widen your table by pulling in related information from a reference table, such as adding product descriptions to a sales table or attaching employee department information to a project list.

Enhancement Activities - Try Left Join and compare results

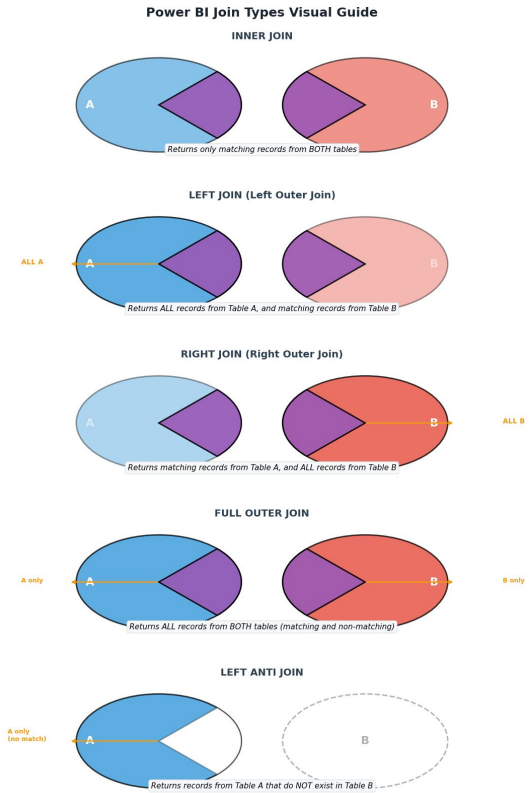




Enhancement Activities - Identify unmatched records.

No unmatched records.

Enhancement Activities - Create a diagram showing join types.



Enhancement Activities - Create a diagram showing join types.

The diagram illustrates five common join types used in Power Query Merge operations:

1. INNER JOIN — Only the overlapping purple section (matching records from both tables)
2. LEFT JOIN — All of Table A (blue) plus matching records from Table B (purple overlap)
3. RIGHT JOIN — Matching records from Table A (purple overlap) plus all of Table B (red)
4. FULL OUTER JOIN — Everything from both tables (all blue, all red, and purple overlap)
5. LEFT ANTI JOIN — Only Table A records that do NOT match Table B (blue crescent only)

Each diagram uses color coding:
Blue = Table A
Red/Orange = Table B
Purple = Matching records (intersection)
Dashed outline = Excluded/unmatched data